

SECTION 9.0

UROLOGY

SURGEON

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DIAGNOSTIC TRANSURETHRAL (ENDOSCOPIC) PROCEDURES

SURGICAL CONSIDERATIONS

Description

Many urologic diseases are diagnosed and evaluated endoscopically through the urethra with the use of specialized instruments, such as cystoscopes and resectoscopes. With the patient in a lithotomy position, the cystoscope is introduced into the urethra and advanced under direct vision all the way into the bladder (Figs. 9.1 and 9.2), allowing inspection of the urethra (**urethroscopy**) and bladder (**cystoscopy**). If pathology is noted, a biopsy can be obtained easily through the cystoscope. It is also possible to introduce a small catheter into the ureteral orifice and advance it up to the kidney for radiologic evaluation (**retrograde pyelography**), to collect a urine specimen, or to bypass areas of obstruction. If the upper urinary tract needs to be visualized, a ureteroscope is introduced through the urethra into the bladder and through the ureteral orifice into the ureter and advanced up to the kidney, allowing inspection of the ureter (**ureteroscopy**) and intrarenal collecting system (**nephroscopy**). These procedures often precede a major surgical operation.

Usual preop diagnosis

Hematuria; hydronephrosis; benign prostatic hypertrophy; cancer of the urethra, prostate, bladder, ureter, and renal pelvis; urinary tract stones; strictures; ureteropelvic junction obstruction; hemorrhagic or interstitial cystitis

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, discussion of carb loading, postop pain management options, and possibility of urinary drain postop. See [Appendix K: Enhanced Recovery After Surgery, p. K-1](#).

■ SUMMARY OF PROCEDURES

	Urethroscopy/Cystoscopy	Ureteroscopy/Nephroscopy
Position	Lithotomy	⇐
Incision	None	⇐
Special instrumentation	Cystoscope	Ureteroscope

Unique considerations	Use of x-ray and fluoroscopy	⇐
Antibiotics	Cefazolin 2 g iv	⇐
Surgical time	15 min	45 min
EBL	None	⇐
Postop care	PACU → home	⇐
Mortality	Minimal	⇐
Morbidity	Infection: 5%	⇐ Ureteral perforation: <5%
Pain score	1	1

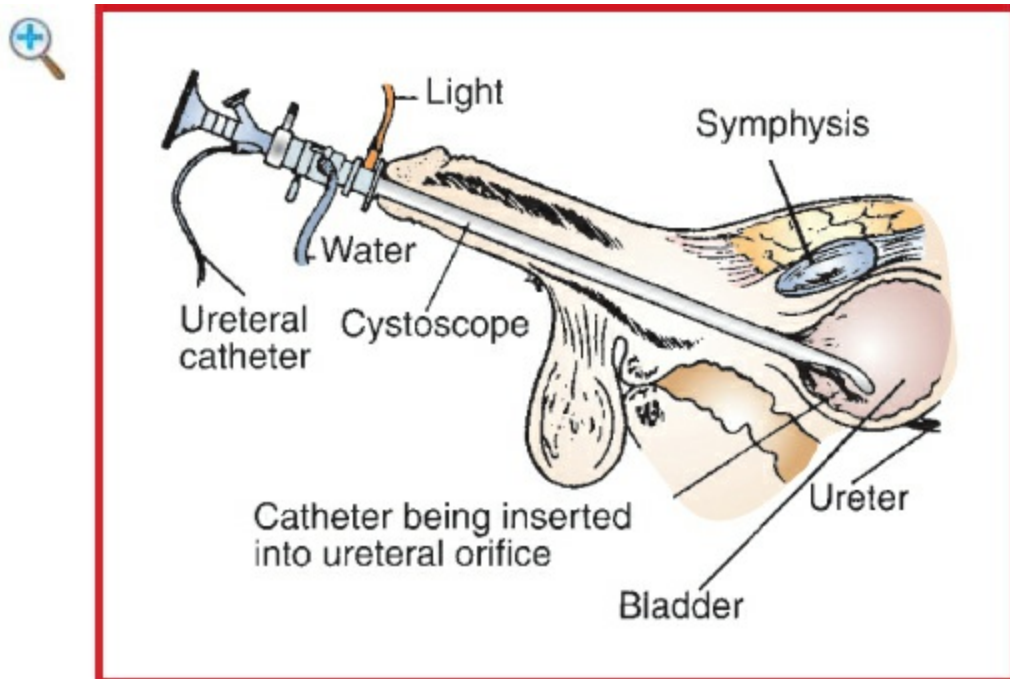


Figure 9.1 Cystoscope introduced into the bladder via the urethra (male anatomy). (Reproduced with permission from Hardy JD: Textbook of Surgery, 2nd edition. JB Lippincott, Philadelphia: 1988.)

■ PATIENT POPULATION CHARACTERISTICS

Age range	All ages	⇐
Male:Female	1:1	⇐
Incidence	30% of all urologic procedures	⇐

Etiology	Hematuria	⇐
	Urethral and bladder tumors	⇐
	Stones	⇐
	Urethral strictures	⇐
	Ureteropelvic junction obstruction	
Associated conditions	Prostatic hypertrophy	Hydronephrosis
	Cystitis	⇐

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations following Therapeutic Transurethral Procedures (Except TURP).

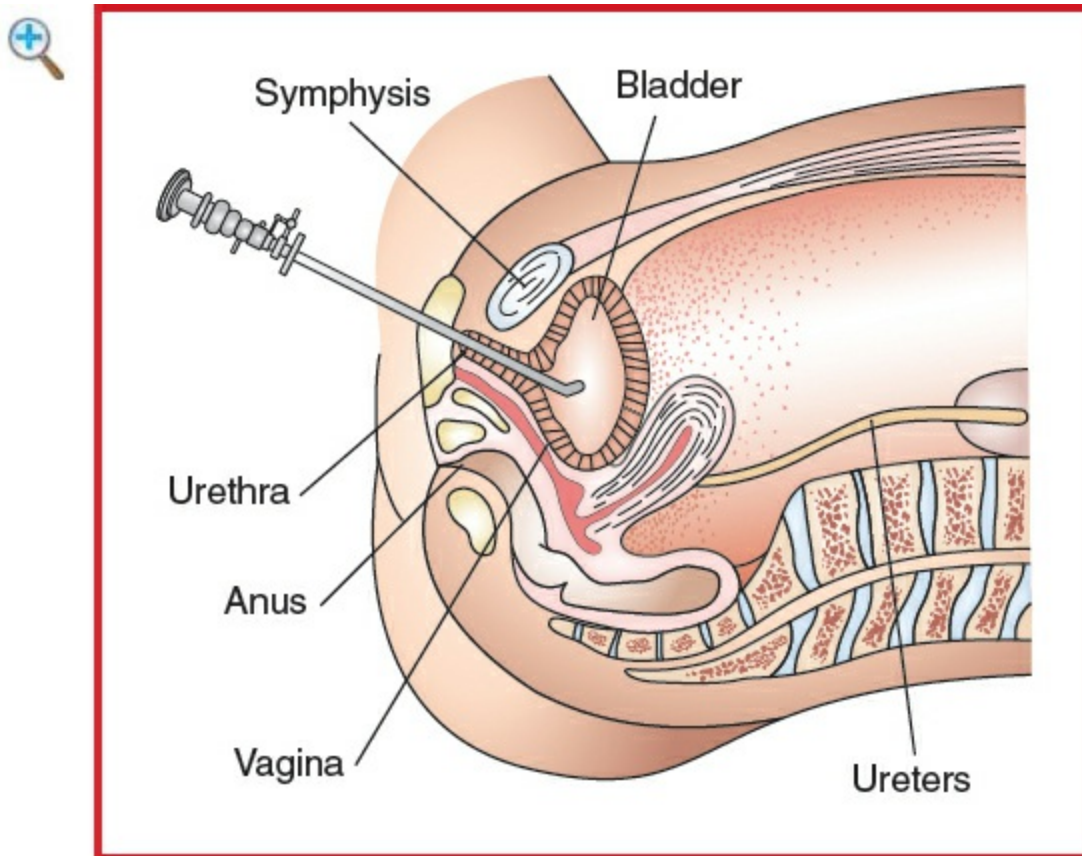


Figure 9.2 Cystoscope introduced into the bladder (female anatomy). (Reproduced with permission from Govan DE: Roche Manual of Urologic Procedures. Hoffmann-La Roche: 1976.)

THERAPEUTIC TRANSURETHRAL PROCEDURES (EXCEPT TURP)

SURGICAL CONSIDERATIONS

Description

Therapeutic transurethral procedures, the most common urologic operations, require the use of specialized instruments, such as cystoscopes and resectoscopes. Because of continuously improving instrumentation and fiberoptics, the range and complexity of these operations are widening, and more operations are being done transurethral now than ever before. These operations are **transurethral resection (TUR)** of any urethral, prostatic, or bladder pathology; **fulguration** of bleeding vessels; **instillation of chemicals**, such as oxychlorosene (Clorpactin) and formalin, into the bladder; **extraction** of stones; and **incision and dilation** of strictures.

With the patient in the lithotomy position, the cystoscope or resectoscope is introduced into the urethra and advanced under direct vision into the bladder, allowing inspection of the urethra and bladder (Figs. 9.1 and 9.2). The pathology is identified. If it is a tumor, it is resected piecemeal with the electrode of the resectoscope, using the cutting current and cauterizing the base of the tumor with the coagulating current. If the pathology is a stone, it is extracted with special forceps or a stone basket. Large stones have to be fragmented, prior to extraction, with a mechanical lithotrite, electrohydraulic probe, ultrasound lithotrite, or laser (holmium or pulsed dye). Chemicals can be instilled through the cystoscope to control interstitial and hemorrhagic cystitis. Bleeding vessels can be coagulated with the electrode. Strictures of the urethra can be dilated or incised with an endoscopic knife. Strictures of the ureter also can be treated endoscopically by dilatation with a balloon catheter or by incision with electrocautery. Balloon catheters with attached cutting electrowires (Acucise) often are used. A temporary ureteral stent is placed at the end of most endoscopic ureteral surgeries.

Variant procedure or approaches

Occasionally, access to the intrarenal collecting system (renal pelvis and calyces) and upper ureter is easier and more appropriately done by a **percutaneous nephrostomy** than by a transurethral procedure. The patient is placed in a prone or flank position, a percutaneous stab wound is made at the costovertebral angle, and a tube is introduced into the kidney under fluoroscopic control.

Usual preop diagnosis

Tumors of the urinary tract; stones; interstitial or hemorrhagic cystitis; strictures

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, discussion of carb loading, postop

pain management options, and possibility of urinary drain postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES		
	Transurethral	Percutaneous
Position	Lithotomy	Flank or prone
Incision	None	Stab wound
Special instrumentation	Cystoscope; resectoscope; catheters; stents	Percutaneous nephrostomy kit; nephroscope; urethroscope; catheters; stents
Unique considerations	Use of x-rays, fluoroscopy, and electrocautery	⇐
Antibiotics	Cefazolin 2 g iv	⇐
Surgical time	1 h	2–3 h
EBL	100 mL	500 mL
Postop care	Irrigation of tubes and catheters to clear clots and prevent obstruction	⇐
Mortality	<1%	⇐
Morbidity	Bleeding: 10%	⇐
	Infection: 5%	⇐
	Perforation: 2%	⇐
	Retained stones: 2%	⇐
Pain score	1	3

■ PATIENT POPULATION CHARACTERISTICS		
Age range	All ages	⇐
Male:Female	1:1	⇐
Incidence	10% of urologic diseases	⇐

Etiology	involve these procedures	
	Urethral and bladder tumors; bladder and ureteral stones; interstitial cystitis; hemorrhagic cystitis; urethral stricture	Kidney stones; upper ureteral stones; ureteropelvic junction obstruction

ANESTHETIC CONSIDERATIONS FOR TRANSURETHRAL PROCEDURES (EXCEPT TURP)

PREOPERATIVE

Patients of all ages may present for ureteral stone extraction. Paraplegics and quadriplegics have a predilection for nephrolithiasis and may present for repeated cystoscopies. Bladder tumors usually are seen in older patients, who may present for cystoscopy or TUR. These patients may have preexisting medical problems, including CAD, CHF, PVD, cerebrovascular diseases, COPD, and/or renal impairment. Preo evaluation should be directed toward the detection and treatment of these conditions prior to anesthesia.

Neurologic	Paraplegics and quadriplegics may present for repeated cystoscopies and stone extractions. Note Hx of autonomic hyperreflexia; Sx may include flushing, headache, and nasal stuffiness, associated with voiding or noxious stimuli below the level of spinal cord injury (see below).
Musculoskeletal	Contractures and pressure sores may make positioning difficult in paraplegics or quadriplegics.
Laboratory	Tests as indicated from H&P
Premedication	Sedation prn anxiety (e.g., lorazepam 1–2 mg po 1–2 h before surgery; midazolam 1–2 mg iv in preop area)
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status.

INTRAOPERATIVE

Anesthetic technique

Spinal, continuous lumbar epidural, and GA are acceptable, with the choice dependent on type and length of procedure, age, coexisting disease, and patient preference. Simpler transurethral procedures (e.g., cystoscopy) are amenable to topical anesthesia, whereas longer and more complex procedures (e.g., ureteral stone extraction) will require regional or GA (see discussion below regarding autonomic hyperreflexia). Note that many of these procedures are done on an outpatient basis, and the anesthetic should be planned accordingly. For regional anesthesia, a sacral block is required for urethral procedures (T9-T10 level for procedures involving the bladder and as high as T8 for procedures involving the ureters).

Regional anesthesia

Topical	2% lidocaine jelly
Spinal	0.75% bupivacaine 10–12 mg. For shorter procedures (<1 h), consider low-dose bupivacaine (0.75%, 7.5 mg), mepivacaine (1.5%, 45 mg), or procaine (10%, 100–150 mg). Lidocaine may be used, but the incidence of transient neurologic symptoms may be as high as 30% for procedures performed in the lithotomy position.
Lumbar epidural	1.5–2.0% lidocaine with epinephrine 5 mcg/mL, 15–25 mL; supplement with 5–10 mL boluses as needed. Supplemental iv sedation

General anesthesia

Induction	Standard induction. ET intubation may not be necessary for shorter procedures; consider LMA. Succinylcholine should be avoided in paralyzed (e.g., paraplegic, quadriplegic) patients 2° $\uparrow$ K ⁺ $\rightarrow$ VF or asystole.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.
Maintenance	Pure inhalation anesthetic (e.g., N ₂ O, sevoflurane/desflurane) for short cases. IV technique

	(e.g., propofol 100–200 mcg/kg/min; supplement with N ₂ O ± volatile anesthetic ± remifentanyl). Muscle relaxation is not essential. Long-acting narcotics are unnecessary because postop pain is usually minimal.	
Emergence	No specific considerations	
Blood and fluid requirements	Usually minimal blood loss IV: 18 ga × 1 BSS @ 1–3 mL/kg/h	See Zero Balance Fluid Management, Appendix K, p. K-2 and p. K-4 .
Monitoring	Standard monitors	
Positioning	✓ and pad pressure points ✓ eyes	NB: In lithotomy position, peroneal nerve compression at lateral fibular head → foot drop.
Complications	Anticipate ↓ BP upon returning from lithotomy position. Autonomic hyperreflexia (AH): • Severe HTN • Bradycardia • Dysrhythmias • Cardiac arrest	Rx: volume (200–500 mL BSS) or ephedrine (5 mg iv) may be necessary. Patients with spinal cord injury level above T10 are at risk for AH associated with stimulation below the level of transection. Transection levels below T5 may be associated with less severe manifestations. AH can be prevented by GA, spinal, or epidural anesthesia. If AH occurs intraop, it should be treated by deepening the level of anesthesia and iv antihypertensive agents (e.g., SNP 0.5–5 mcg/kg/min; labetalol 5–10 mg iv; phentolamine

2–5 mg iv), if necessary.

▼ POSTOPERATIVE

Complications	Peroneal nerve injury 2° lithotomy position	Peroneal nerve injury manifested as foot drop with loss of sensation over the dorsum of the foot. Seek neurology consultation. Bladder perforation may present as shoulder pain in the awake patient but may go unnoticed in a patient under GA. Sx include unexplained HTN, tachycardia, ↓ BP (rare).
	Bladder perforation	
	Fever/bacteremia	
Multimodal pain management	Pain usually mild	Rx: acetaminophen 1000 mg, fentanyl 25–50 mcg iv, ketorolac 15–30 mg im or iv
ERAS	PONV protocol (see Appendix B, p. B-4 and p. B-6)	Encourage early enteral nutrition, Foley catheter removal, and mobilization.

Suggested Readings

1. Cockcroft JO, Berry CB, McGrath JS, et al: Anesthesia for major urologic surgery. *Anesthesiol Clin* 2015; 33(1):165–72.
2. Liu N, Zhou M, Biering-Sørensen F, et al: Iatrogenic urological triggers of autonomic dysreflexia: a systematic review. *Spinal Cord* 2015; 53(7):500–9.
3. Vukovic N, Dinic L: Enhanced recovery after surgery protocols in major urologic surgery. *Front Med (Lausanne)* 2018; 5:93.

TRANSURETHRAL RESECTION OF THE PROSTATE (TURP)

SURGICAL CONSIDERATIONS

Description

TURP is one of the most common urologic operations, performed to relieve bladder outlet obstruction by an enlarging prostate gland. It is often preceded by **cystoscopy**, which is used to evaluate the size of the prostate gland and to rule out any other pathology, such as bladder tumor or stone. The operation is performed with the resectoscope, a specialized instrument having an electrode capable of transmitting both cutting and coagulating currents. Resectoscopes are either single inflow only or continuous flow with an inflow and outflow system. The latter allows the surgeon to maintain low pressure in the bladder and prostatic fossa and thus limit fluid absorption. Two different electrical options are available. The traditional resectoscope is a monopolar system, and this requires a grounding pad and possible interference with electric devices, such as pacemakers. Bipolar resectoscopes have both the active and return electrode fitted in the resectoscope and do not cause any electrical interference with cardiac electrical devices. In addition, the use of bipolar cautery allows saline to be used as an irrigant during surgery.

The resectoscope is introduced into the bladder ([Fig. 9.3](#)), and the tissue protruding into the prostatic urethra is resected in small pieces called “chips.” Bleeding vessels are coagulated with the coagulating current. The resection is performed with continuous irrigation using an isotonic solution, such as sorbitol 2.7% with mannitol 0.54% in monopolar resectoscopes and normal saline in bipolar resectoscopes. After the obstructing prostatic tissues are completely resected and bleeding vessels coagulated, the chips are irrigated from the bladder and the resectoscope is removed. An indwelling Foley catheter is introduced into the bladder. The time of transurethral resection should not exceed 2 h because excessive absorption of the irrigating fluid may → dilutional hyponatremia, confusion, seizures, and heart failure. However, this is less of an issue with a continuous flow bipolar resectoscope where saline is used as an irrigant. Although fluid absorption can occur, hyponatremia does not occur with the use of saline. The size of the enlarged prostate or adenoma, therefore, needs to be carefully assessed preop to determine if it is possible to complete the resection within 2 h. If not, an **open prostatectomy** is performed. This variant approach is discussed under Open Prostate Operations.

Variant procedure or approaches

A number of techniques have been developed to avoid the morbidity of TURP. These are either **vaporization** (electrocautery or laser) or **thermocoagulation** of the prostate (laser, microwave, radiofrequency). The following techniques are available and approved:

TUVP

Transurethral vaporization of the prostate with a standard resectoscope using a roller ball electrode at 275–300 watts setting

Laser ablation (PVP)

Laser coagulation of the prostate was done with Nd:YAG or Ho:YAG laser through a standard cystoscope. This procedure has been largely replaced by photoselective vaporization of the prostate (PVP) with an 80-w KTP laser, 120- to 180-w GreenLight laser, or diode laser. These green spectrum laser wavelengths allow 1–2 mm deep vaporization of prostate tissue with minimal blood loss. This is currently the most popular minimally invasive technique used for treatment of BPH. It can also be done on patients while on anticoagulation or with bleeding disorders. All personnel in the OR, including the patient, must wear protective glasses to protect the eyes from inadvertent exposure from a break in the laser fiber.

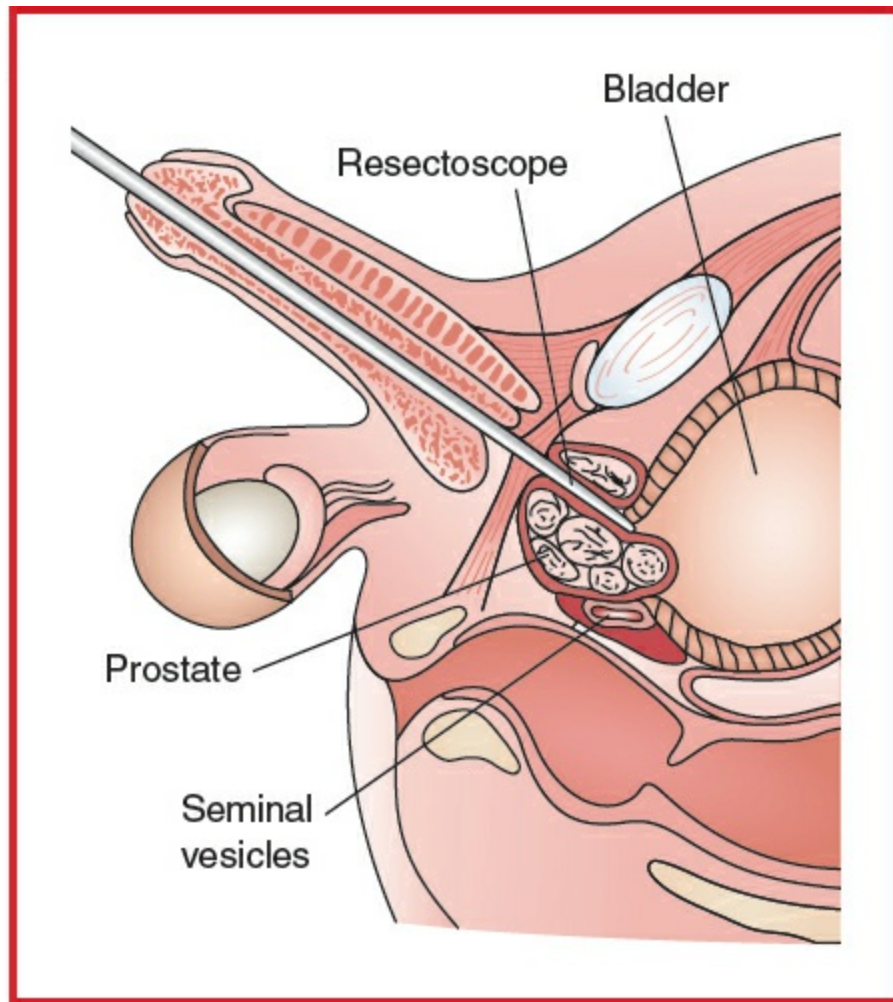


Figure 9.3 Transurethral resection of the prostate using a resectoscope. (Reproduced with permission from Govan DE: Roche Manual of Urologic Procedures. Hoffmann-La Roche: 1976.)

TUNA

Transurethral needle ablation of the prostate is done with a special disposable device connected to a radiofrequency generator.

TUMT

Transurethral microwave thermotherapy is done with a catheter that has a microwave antenna attached to it. A microwave generator is needed for this procedure.

UroLift™

Prostatic urethral lift (UroLift) implants are transmural tethers (see [Fig. 9.4](#)) used to enlarge the urethral lumen. They are placed using a rigid cystoscope, usually with local anesthesia ± sedation.

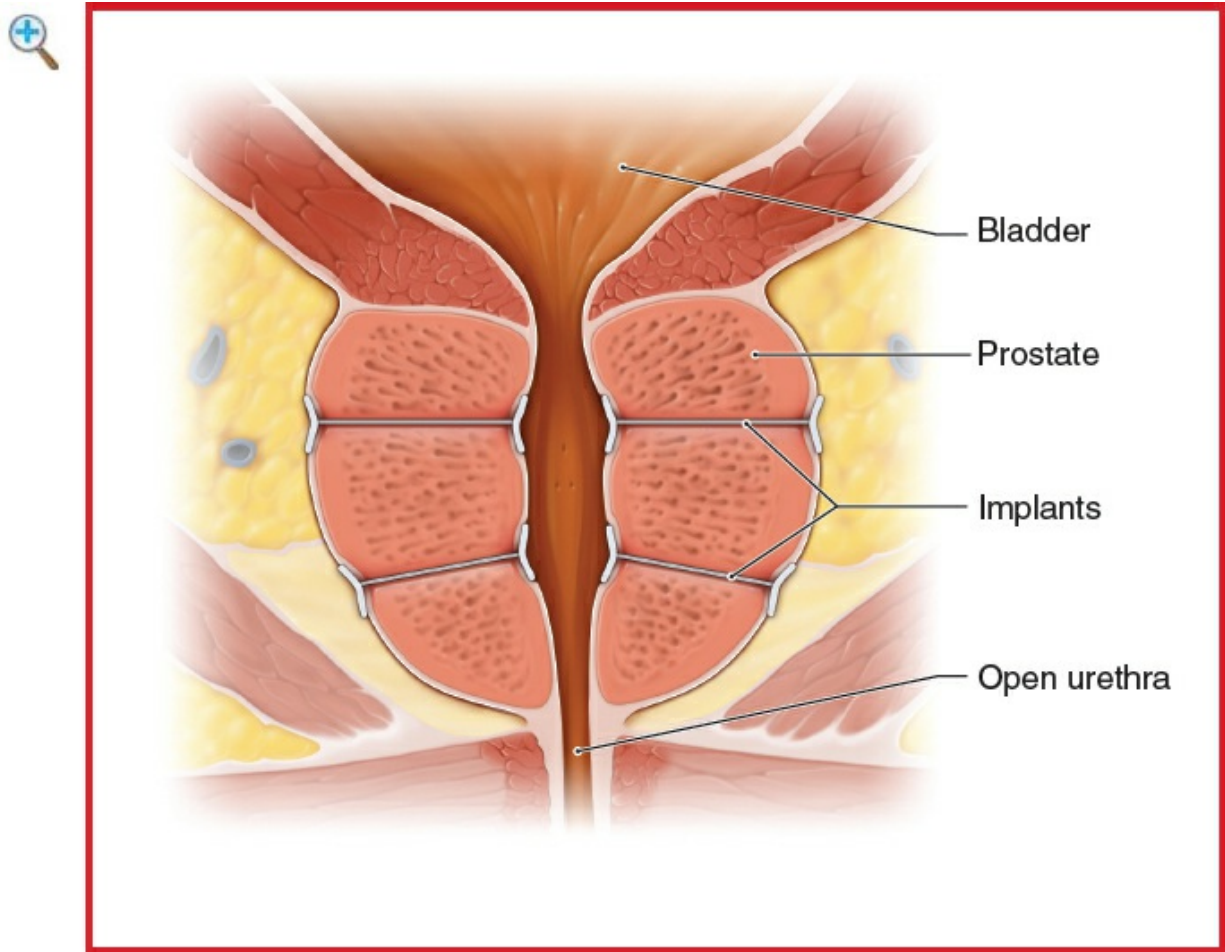


Figure 9.4 Dilated urethra following transurethral placement of UroLift® implants.

All of the above have several advantages over TURP, including shorter surgical time, no blood loss, and reduced risk of fluid absorption and all can be done as outpatient procedures.

Usual preop diagnosis

Benign prostatic hypertrophy; prostate cancer

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, postop pain management options, and possibility of urinary drain postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES

	TURP	Thermotherapy
Position	Lithotomy	←

Incision	None	⇐
Special instrumentation	Cystoscope; resectoscope; catheters; electrocautery	Cystoscope; resectoscope; catheters; electrocautery (standby); laser equipment
Unique considerations	During resection, the patient should be absolutely still because any movement may lead to perforation or injury to the external sphincter, → postop incontinence.	During laser resection, the patient and all personnel should wear protective eyeglasses.
Antibiotics	Cefazolin 2 g iv	⇐
Surgical time	1–2 h (not to exceed 2 h)	1 h
EBL	500 mL	None
Postop care	Irrigation of the Foley catheter to clear it of clots and keep it from being blocked. Determination of serum Na ⁺	⇐
Mortality	<1%	⇐
Morbidity	Significant intraop bleeding: 10% Intraop perforation and fluid extravasation, which may require placement of retropubic drain: <0.5% Postop bleeding, which may necessitate a return to OR for fulguration of bleeding vessels: <5% Absorption of irrigating fluid, which may → dilutional hyponatremia, mental confusion, and	Prolonged catheterization: 10% Postop bleeding: 1%

heart failure: 2–5%

Pain score

1

1

■ PATIENT POPULATION CHARACTERISTICS

Age range	49–90 yr; typically, 70s and 80s
Incidence	Very common; 90% of men will develop benign hypertrophy; 20% may need surgical intervention.
Etiology	Aging; benign prostatic hypertrophy; prostate cancer
Associated conditions	COPD (10%); heart disease (10%); HTN (10%); diabetes mellitus (DM) (5%); DIC (1–2% of patients with prostate cancer may also have a low-grade, subclinical DIC, which becomes clinically manifest postop)

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Patients presenting for prostate surgery are generally older and may have preexisting medical problems, including CAD, CHF, PVD, cerebrovascular disease, COPD, an renal impairment. Preop evaluation should be directed toward the detection and treatment of these conditions before anesthesia.

Respiratory	COPD common in this age group. Patients with >50 pack-year smoking Hx, or with any respiratory Sx, may need PFTs. For dyspnea with moderate exercise, ✓ VC, FEV₁, and MMEF. If VC < 80%, FEV₁ < 60%, or MMEF < 40% is predicted, ✓ ABG. If ABG and PFT are markedly abnormal, consider postponing surgery until patient's respiratory condition has been optimized. Tests: PFT; CXR; ABG, as indicated from H&P
Cardiovascular	HTN, CAD common in this age group. Assess exercise tolerance by H&P (e.g., should be able to climb a flight of stairs without difficulty or SOB). Tests: ECG; others as indicated from H&P

Neurologic	Cerebrovascular disease, Alzheimer's, and other neurologic problems may be present in this age group. Assess mental status to guide evaluation of any intraop or postop changes.
Renal	Anticipate renal impairment 2° chronic obstruction. Tests: BUN; Cr; electrolytes. If ↑ BUN and ↑ Cr, ✓ creatinine clearance (nl = 95–140 mL/min).
Musculoskeletal	Various arthritides in this age group may cause problems with positioning for regional anesthesia and surgery.
Endocrine	Increased incidence of DM
Hematologic	Moderate blood loss expected with larger glands. If the gland <80 g, no T&C is necessary. Test: Hct
Laboratory	Other tests as indicated from H&P
Premedication	Continue commonly used drugs (e.g., digitalis, β-blockers, NTG) to prevent cardiovascular problems. Sedation prn anxiety (e.g., midazolam 1–2 mg iv before surgery)
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40 mg po with sip of water for GA patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

Regional or GA. Choice of technique depends on the coexisting disease and the patient's preference. Regional anesthesia may hold some advantage over GA for TURF in that it allows evaluation of mental status and, thus, earlier detection of TURF syndrome. The incidence of postdural puncture headache is very low in this age group (<1%). A T9 level is optimal. Continuous lumbar epidural anesthesia has no advantage over (and may be less reliable than) spinal anesthesia for TURP. Since the procedure is relatively short, supplemental doses are usually not necessary.

Regional anesthesia

Spinal	0.75% bupivacaine, 12 mg in 7.5% dextrose solution (1.6 mL)
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General anesthesia

Induction	Standard induction	
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.	
Maintenance	Standard maintenance. Muscle relaxation is not mandatory, although patient movement during the procedure must be avoided.	
Emergence	Postop pain is usually not significant. Anticipate ↓ BP when legs are repositioned from lithotomy. Avoid stress on the lumbar spine by slowly and simultaneously bringing legs together and returning to supine position. PONV prophylaxis should include ondansetron.	
Blood and fluid requirements	Moderate blood loss (TURP) Minimal blood loss (thermotherapy) IV: 16–18 ga × 1 BSS @ 1–3 mL/kg/h	Blood loss can be large (TURP) if venous sinuses are entered and difficult to quantify because of irrigant. To flush away blood and tissue and to promote visibility during TURP (or thermotherapy), continuous irrigation is used. For monopolar procedures, irrigating fluid must be nonelectrolytic to prevent dispersion of current but near iso-osmotic to prevent hemolysis. For these reasons, sorbitol

(2.7%) with mannitol (0.54%) or glycine (1.5%) is added to distilled water to produce solutions that are nearly isotonic. A zero balance fluid management strategy is recommended (see [p. K-3](#)). See Zero Balance Fluid Management, [p. K-2](#).

Monitoring

Standard monitors

Regional anesthesia allows monitoring of mental status. Invasive monitoring, if indicated from H&P

TURP syndrome

Intravascular volume overload
Hyponatremia
+Hypotonicity 2°
absorption of irrigant
Symptoms include:

- N/V
- Visual disturbances
- Mental status changes
- Coma
- Sz
- HTN
- Angina
- Cardiovascular collapse

Factors that influence the absorption of irrigant include surgical technique (TURP or thermotherapy), hydrostatic pressure of irrigant (height of bag), number of venous sinuses opened, peripheral venous pressure, duration of surgery, and experience of the surgeon. Resections should optimally be limited to 1 h or less. Some CNS manifestations (e.g., visual disturbances, nausea, confusion) are 2° glycine and its metabolites. Rx may include observation, diuresis (e.g., furosemide 5–20 mg iv), and

		administration of hypertonic saline (e.g., 100 mL of 3% saline over 1–2 h). Serum sodium <120 is associated with more severe symptoms, and the goal of therapy is to restore sodium to >120. In milder cases, observation and water restriction may be sufficient.
Positioning	✓ and pad pressure points ✓ eyes	NB: In lithotomy position, peroneal nerve compression at the lateral fibular head → foot drop
Complications	Bladder perforation TURP syndrome	Bladder perforation may produce shoulder pain in the awake patient. Bladder perforation (and TURP syndrome) may go unnoticed under GA; Sx: ↑ BP, ↑ HR (occasionally ↓ BP).

POSTOPERATIVE

Complications	TURP syndrome Bladder perforation Fever/bacteremia/sepsis Hypothermia	See discussion of TURP syndrome above.
Multimodal pain management	Minimal postop pain	Rx: acetaminophen 1 g is often sufficient.
ERAS	PONV protocol (see Appendix B, p. B-4 and p.	Encourage early enteral nutrition, Foley catheter

	B-6)	removal, and mobilization.
Tests	Hct; electrolytes Blood cultures if febrile	Consider serum osmolality, CXR, ECG in TURP syndrome.

Suggested Readings

1. Cornu JN: Bipolar, monopolar, photovaporization of the prostate, or Holmium Laser enucleation of the prostate: how to choose what's best? *Urol Clin North Am* 2016; 43(3):377–84.
2. Hawary A, Mukhtar K, Sinclair A, et al: Transurethral resection of the prostate syndrome: almost gone but not forgotten. *J Endourol* 2009; 23(12):2013–20.
3. Longo MA, Cavalheiro BT, de Oliveira Filho GR: Systematic review and meta analyses of tranexamic acid use for bleeding reduction in prostate surgery. *J Clin Anesth* 2018; 48:32–8.
4. Magistro G, Stief CG, Gratzke C: Prostatic urethral lift versus transurethral resection of the prostate (TURP). *Curr Urol Rep* 2017; 18(10):82.
5. Smith RD, Patel A: Transurethral resection of the prostate revisited and updated. *Curr Opin Urol* 2011; 21(1):36–41.
6. Teng J, Zhang D, Li Y, et al: Photoselective vaporization with the green light laser vs transurethral resection of the prostate for treating benign prostate hyperplasia: a systematic review and meta-analysis. *BJU Int* 2013; 111(2):312–23.
7. Vukovic N, Dinic L: Enhanced recovery after surgery protocols in major urologic surgery. *Front Med (Lausanne)* 2018; 5:93.
8. Wang J, Zhang C, Tan G, et al: Risk of bleeding complications after preoperative antiplatelet withdrawal versus continuing antiplatelet drugs during transurethral resection of the prostate and prostate puncture biopsy: a systematic review and meta-analysis. *Urol Int* 2012; 89(4):433–8.

OPEN PROSTATE OPERATIONS

SURGICAL CONSIDERATIONS

Description

Open (in contrast to transurethral or endoscopic) operations on the prostate gland are less common. They include **simple prostatectomy** and **radical prostatectomy** and are done through either a midline extraperitoneal incision, which extends from the umbilicus

to the symphysis pubis, or through a Pfannenstiel incision (Fig. 9.5, inset).

Simple prostatectomy

When the benign prostatic hypertrophy or adenoma is too large to be resected transurethrally, it is removed by a simple prostatectomy. The prostate gland is exposed through a retropubic approach (Fig. 9.5A), and the anterior capsule is incised, exposing the adenoma—the central part of the prostate—which is excised, “shelled” out by blunt dissection (Fig. 9.5B), leaving behind the peripheral prostate and all the associated structures. A Foley catheter is left indwelling in the urethra, and the incision in the prostate capsule is closed. In a **suprapubic prostatectomy**, the incision is made in the bladder and the adenoma shelled from within the bladder.

Radical prostatectomy

The term **radical prostatectomy** is used because the entire prostate, both seminal vesicles and pelvic nodes, is removed. It is used to differentiate this cancer operation from a simple prostatectomy (used for benign prostatic hypertrophy). Radical prostatectomy can be achieved through a **retropubic approach**. In radical prostatectomy, all of the prostate gland is removed, together with the bladder neck, the seminal vesicles, and the ampullae of the vas deferens. A **limited pelvic lymphadenectomy** (Fig. 9.6) also is performed. After the prostate gland and its associated structures are removed, the bladder neck is reduced to 1 cm diameter and anastomosed to the membranous urethra over an indwelling Foley catheter. Most of the blood loss occurs during control of the dorsal vein complex. In early-stage cancers, which nowadays comprise 90% of patients, a nerve sparing procedure is done. This involves preserving potency by preserving the nerves to the corpora cavernosa. More recently, radical prostatectomy is being done by laparoscopy. This procedure has been popularized with the use of robots. Please see section of robotic urologic surgery.

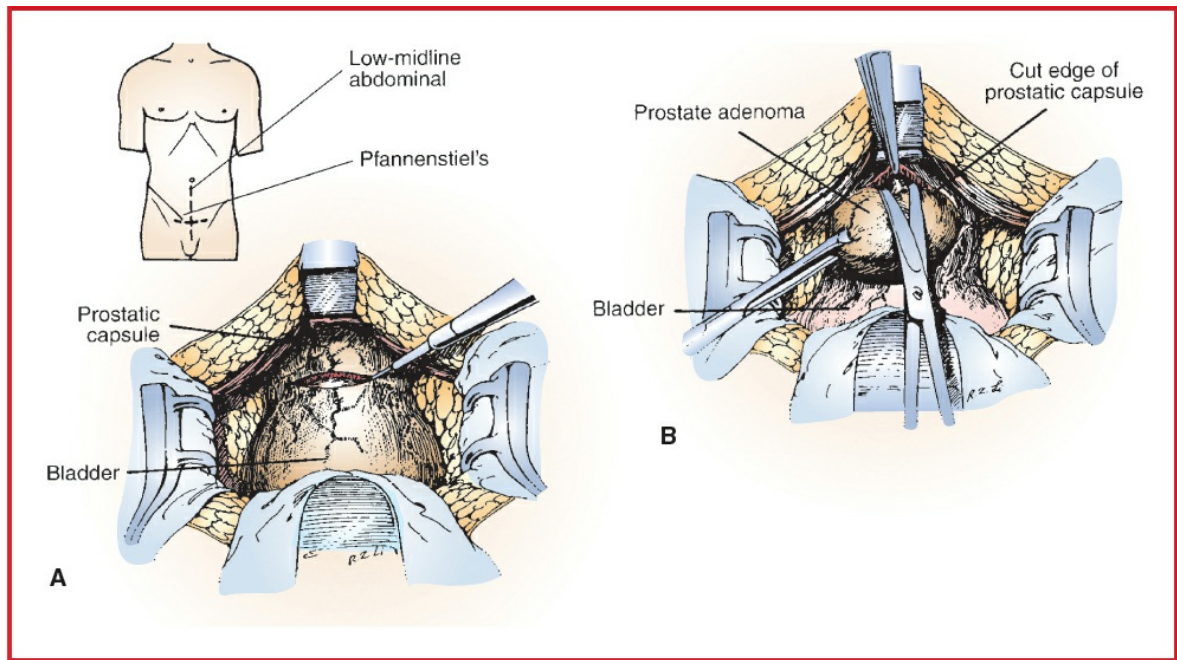


Figure 9.5 Retropubic prostatectomy. **A:** A transverse capsulotomy is made between heavy hemostatic stay sutures. **B:** The cleavage plane between the adenoma and the surgical capsule is developed with scissors. (Reproduced with permission from Fowler JE: Manual of Urologic Surgery Little, Brown, Boston: 1990.)

Variant approach

Minimally invasive prostate cancer surgery has expanded with the use of the **da Vinci robotic system**. Most laparoscopic radical prostatectomies are done with the assistance of the robot. With the robotic system, the surgeon sits at a console away from the patient and has a three-dimensional view with 10× magnification. A bedside assistant is necessary for instrument change and retraction. See [page 1022](#) for special Anesthetic Considerations for Robotic-Assisted Laparoscopic Surgery.

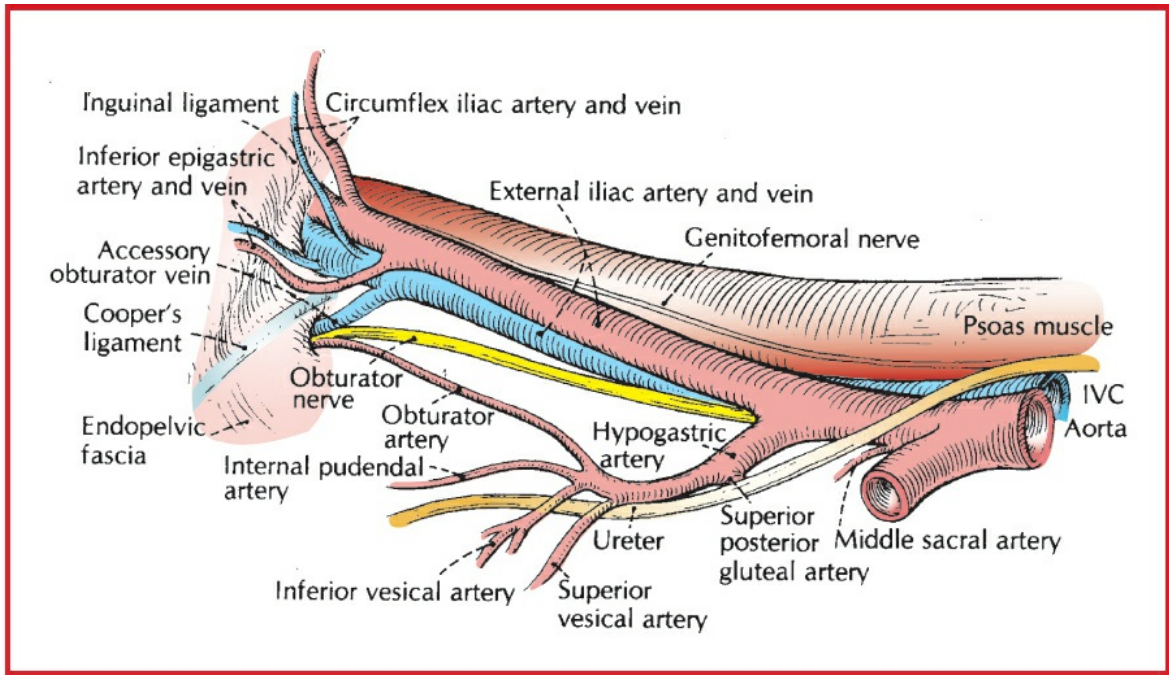


Figure 9.6 Right lateral pelvic wall. Anatomy of pelvic blood vessels and nerves encountered in a pelvic lymph node dissection. (Reproduced with permission from Graham SD Jr. Glenn's Urologic Surgery, 5th edition. Lippincott-Raven, Philadelphia: 1998.)

Usual preop diagnosis

Benign prostatic hypertrophy; prostate cancer

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, postop pain management options, postop recovery plan, and urinary drain postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES

	Simple Prostatectomy	Radical-Retropubic
Position	Supine	⇐; lithotomy → steep Trendelenburg (robotic)
Incision	Extraperitoneal, low midline (Fig. 9.7), or Pfannenstiel's (Fig. 9.5A , inset)	⇐; multiple port sites (robotic)
Unique considerations	None	⇐; da Vinci robotic system is used increasingly (>80%); abdominal

insufflation.

Antibiotics	Cefazolin 2 g iv	Cefazolin 2 g iv
Surgical time	1 h	2–3 h (experienced surgeon)
EBL	500 mL	1500 mL (less with robotic assist)
Postop care	Irrigate catheter to clear blood clots and prevent obstruction, frequently, if urine is bloody.	⇐
Mortality	<1%	⇐
Morbidity	Bleeding: 2% DVT: 2% Infection: 2% PE: 1%	⇐ ⇐ ⇐ ⇐ Impotence: Non–nerve-sparing: 100% Nerve-sparing: 20–50% Lymphocele: 4%
Pain score	8	8

■ PATIENT POPULATION CHARACTERISTICS

Age range	40–80 yr
Incidence	20% of men will develop symptomatic benign prostatic hypertrophy; 9% will develop clinically evident prostate cancer.
Etiology	Aging
Associated conditions	COPD (10%); CAD (10%); HTN (10%); diabetes mellitus (5%); renal failure (1%)

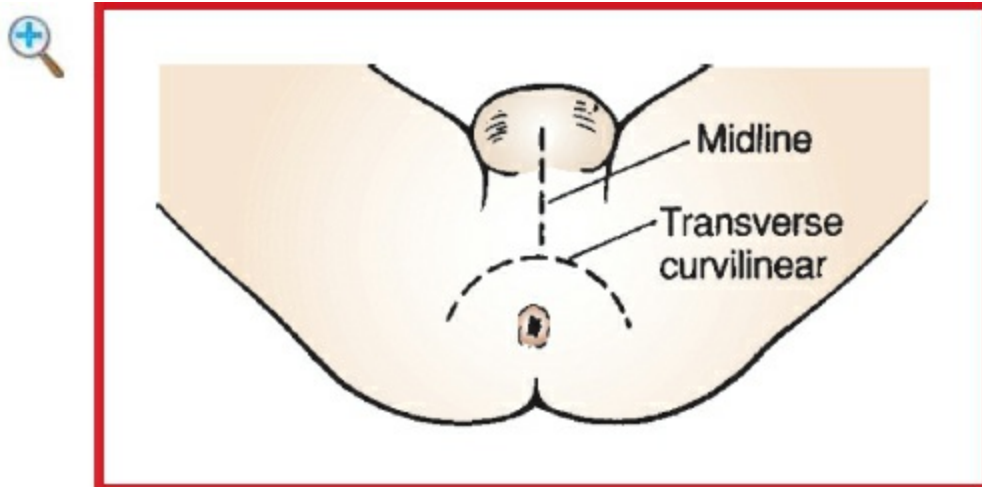


Figure 9.7 Perineal incisions.

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Patients presenting for prostate surgery are generally elderly and may have preexisting medical problems, including CAD, CHF, PVD, cerebrovascular disease, COPD, and renal impairment. Preop evaluation should be directed toward the detection and treatment of these conditions prior to anesthesia. A bowel prep may have been completed before surgery.

Respiratory

COPD common in this age group. Patients with Hx of >50 pack-year smoking, or with respiratory Sx, may require PFTs. For dyspnea with moderate exercise, check VC, FEV₁, and MMEF. If VC < 80%, FEV₁ < 60%, or MMEF < 40% is predicted, ✓ ABG. If ABG and PFT are markedly abnormal, consider postponing surgery until patient's respiratory condition has been optimized. In robotic surgery, steep Trendelenburg position will compromise pulmonary function.

Tests: PFT; CXR; ABG, as indicated from H&P

Cardiovascular

HTN, CAD common in this age group. Assess exercise tolerance by H&P (e.g., should be able to climb a flight of stairs without difficulty or SOB). Transition plans for patients on anticoagulants for coronary stents should be discussed with both the cardiologist and surgeon before surgery.

	Tests: ECG; others as indicated from H&P
Neurologic	Cerebrovascular disease, Alzheimer's, and other neurologic problems may be present in this age group. Assess mental status to guide evaluation of any intraop or postop changes. In robotic cases, steep Trendelenburg position and CO ₂ insufflation → ↑ ICP and ↑ intraocular pressure. A preop neurosurgical consult is recommended for patients with intracranial lesions or VP shunts. Blindness has been reported following robotic prostatectomy.
Renal	Anticipate renal impairment 2° chronic obstruction. Pneumoperitoneum during robotic surgery may ↓ renal blood flow and oliguria postop. Test: Cr
Musculoskeletal	Various arthritides may cause problems with positioning for regional anesthesia and surgery.
Endocrine	Increased incidence of diabetes mellitus
Hematologic	Moderate blood loss expected with larger glands. For glands <30 g, no T&C is necessary; for glands 30–80 g, T&C 2 U PRBCs; for glands >80 g, T&C 4 U PRBCs. Test: Hct
Laboratory	Other tests as indicated from H&P
Premedication	Continue commonly used drugs (e.g., digitalis, β-blockers, diuretics, NTG) to prevent cardiovascular complications. Sedation prn anxiety (e.g., midazolam 1–2 mg iv on call to OR) If ERAS protocol is followed, consider preoperative acetaminophen 1000 mg po and/or gabapentin 600–1200 mg po or pregabalin 150–300 mg po 30–60 min prior to induction.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of a rectus sheath catheter or epidural block for postop pain management. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40 mg

po with sip of water for patients with h/o severe PONV.

◆ INTRAOPERATIVE

Anesthetic technique

Regional (spinal, continuous spinal, continuous lumbar epidural), GA, or combined techniques are acceptable for open procedures. Robotic procedure typically requires GA. If regional anesthesia is used, optimal block level is T8-T10 (depending on the incision site). Advantages of regional anesthesia include potential for lower intraop blood loss, possible lower incidence of DVT postop, and faster return of bowel function. Disadvantages include positioning considerations (see below).

Regional anesthesia

Spinal	Bupivacaine (0.75%) 12 mg in 7.5% dextrose solution (1.6 mL)
Epidural	1.5–2% lidocaine with epinephrine 5 mcg/mL, 15–25 mL, supplemental iv sedation as necessary. Additional epidural lidocaine (5–10 mL boluses) may be needed, depending on the length of procedure. Alternatively, smaller volumes of bupivacaine (0.25–0.5%) or ropivacaine (0.5–1%) may be used. The addition of opiates has been associated with ↑ urinary retention in noncatheterized patients.

General anesthesia

Induction	Standard induction
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider preincision placement of nerve blocks.
Maintenance	Standard maintenance. In robotic cases, recheck ETT position after the patient has been positioned and the abdomen insufflated to avoid mainstem intubation. Patient movement is not well tolerated by the robot → visceral/vascular injury. Consider

	ketamine 0.5 mg/kg bolus + 0.25 mg/kg/h infusion during surgery (especially in opioid-dependent patients).	
Emergence	No special considerations. Airway edema may occur after prolonged steep Trendelenburg position. Discuss with surgeon local wound infiltration or nerve block placement or reinforcement if initially placed >4 h ago. Consider iv acetaminophen if po dose is given >6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.	
Blood and fluid requirements	Moderate-to-large blood loss IV: 14–16 ga × 1–2 BSS @ 1–3 mL/kg/h	Additional requirements dependent on the type of anesthesia. Regional techniques may require the use of vasopressors or volume (if indicated) to offset sympathectomy and systemic vasodilation; RA also may be associated with lower blood loss than GA. A zero balance fluid management strategy may be beneficial (see Appendix K, p. K-2 and p. K-4.). See Goal-Directed Fluid Management, Appendix K, p. K-4.
Monitoring	Standard monitors Depending on underlying disease: ± CVP ± Arterial line	Arterial line is often useful for continuous BP and PPV monitoring. Patients at particularly high risk (e.g., Hx of preexisting cardiopulmonary disease) may benefit from both ART and CVP lines.

Positioning	Anticipate ↓ BP on return from lithotomy position. ✓ and pad pressure points ✓ eyes	Rx: ephedrine (5–10 mg iv) or volume (250 mL BSS) may be necessary. Elderly patients with arthritis or respiratory impairment may not tolerate the extreme positioning associated with robotic prostatectomy. NB: In lithotomy position, peroneal nerve compression at the lateral fibular head → foot drop.
Complications	Indigo carmine reaction Hemorrhage Hypothermia VAE Robot malfunction	Indigo carmine → false ↓ O₂ sat ± ↑ BP; rare allergic reaction → rash + bronchoconstriction + ↓ BP Insufflation pressure may exceed venous pressure → VAE.
ERAS	PONV protocol (see Appendix B, p. B-4 and p. B-6)	Encourage early enteral nutrition, Foley catheter removal, and mobilization. Consider DVT prophylaxis. Chewing gum promotes return of bowel function.

POSTOPERATIVE

Complications	Peroneal nerve injury 2° lithotomy position DVT Airway edema → obstruction	Manifested by foot drop with loss of sensation on the dorsum of the foot. Seek neurology consultation.
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		Incidence of DVT less with regional than GA Sx: variable, with pain and tenderness over involved area
Multimodal pain management	Significant postop pain. Rx: hydromorphone 0.01–0.02 mg/kg iv in incremental doses (e.g., 10–15 min prn) or PCA	Emphasize multimodal approach including regional analgesia, acetaminophen, ± celecoxib, and ± gabapentin. Consider epidural infusion of dilute local anesthetics/opiates or PCA. Robotic procedure patients have lower postop pain.
ERAS	PONV protocol (see Appendix B, p. B-4 and p. B-6)	Encourage early enteral nutrition, Foley catheter removal, and mobilization. Consider DVT prophylaxis. Chewing gum promotes return of bowel function.
Tests	Hct	

Suggested Readings

1. Basiri A, de la Rosette JJ, Tabatabaei S, et al: Comparison of retropubic laparoscopic and robotic radical prostatectomy: who is the winner? *World J Urol* 2018; 36(4):609–21.
2. Cockcroft JO, Berry CB, McGrath JS, et al: Anesthesia for major urologic surgery. *Anesthesiol Clin* 2015; 33(1):165–72.
3. Kalmar AF, Dewaele F, Foubert L, et al: Cerebral haemodynamic physiology during steep Trendelenburg position and CO₂ pneumoperitoneum. *Br J Anaesth* 2012; 108(3):478–84.
4. Vukovic N, Dinic L: Enhanced recovery after surgery protocols in major urologic surgery. *Front Med (Lausanne)* 2018; 5:93.
5. Whalley DG, Berrigan MJ: Anesthesia for radical prostatectomy, cystectomy, nephrectomy, pheochromocytoma, and laparoscopic procedures. *Anesthesiol Clin*

NEPHRECTOMY

SURGICAL CONSIDERATIONS

Description

Nephrectomies fall into three basic groups: simple, partial, and radical. (Surgical anatomy is shown in [Fig. 9.8](#).)

Simple nephrectomy, performed for benign conditions, is the surgical excision of the kidney and a small segment of proximal ureter. The dorsal approach is well suited for this operation and begins with an incision extending from the 12th rib to the iliac crest along the lateral edge of the sacrospinalis muscle and quadratus lumborum muscle. The dorsolumbar fascia is opened, exposing Gerota fascia and the perinephric fat ([Fig. 9.9](#)). The kidney is mobilized until the hilum is exposed. The artery and vein are tied, suture-ligated, and transected. The ureter is followed distally as far as possible, tied, and transected. The kidney is delivered out of the incision, which is then closed by approximating the dorsolumbar fascia and the fascia of the sacrospinalis muscle.

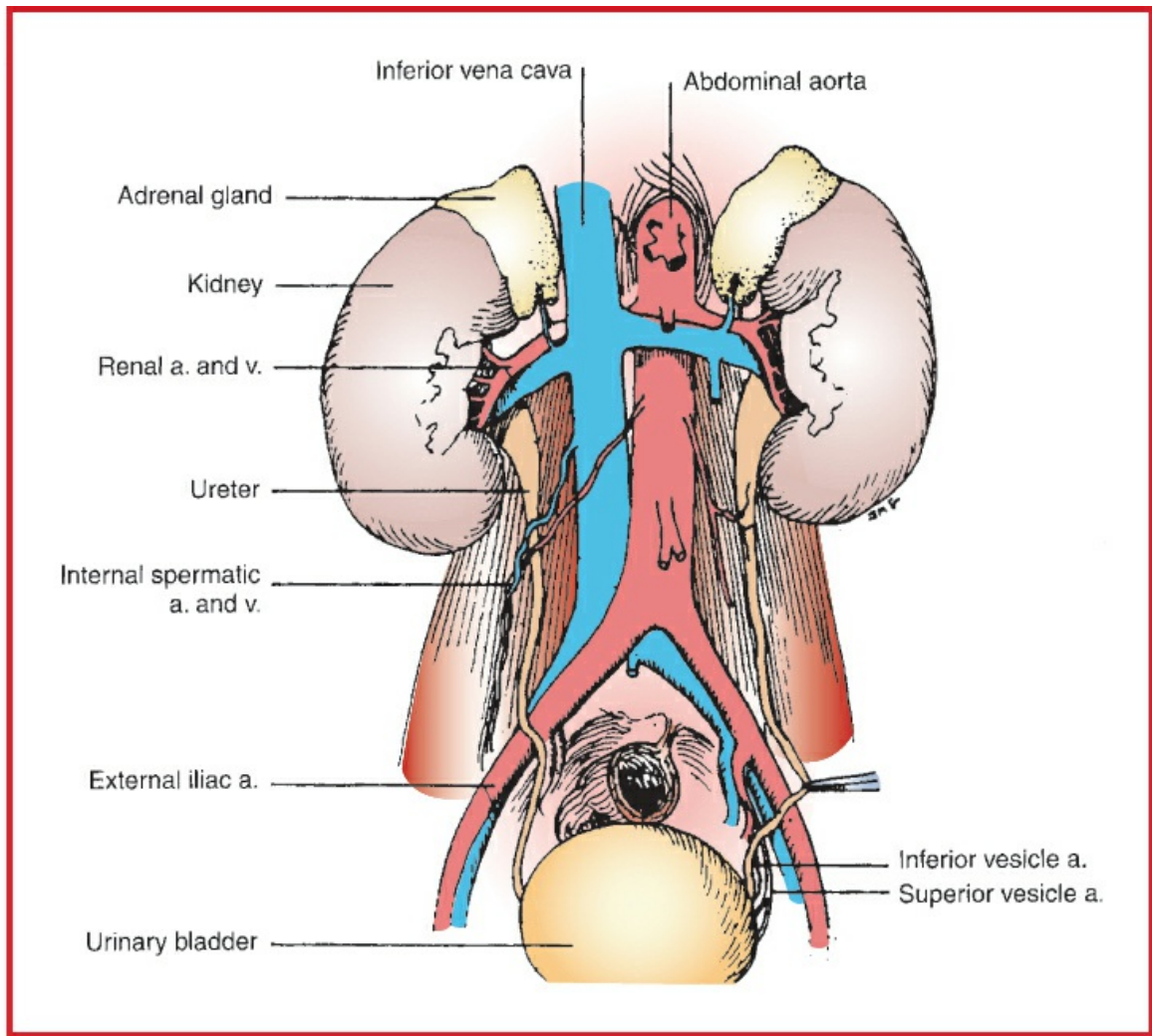


Figure 9.8 Surgical anatomy of the urinary tract. (Reproduced with permission from Hardy JD: Hardy's Textbook of Surgery, 2nd edition. JB Lippincott, Philadelphia: 1988.)

Usual preop diagnosis

Chronic hydronephrosis; hypoplastic kidney; renovascular HTN; double collecting system

Partial nephrectomy is the surgical excision of the segment of the kidney harboring the pathology. It is performed for small renal cell carcinomas and benign tumors of the kidney, such as angiomyolipomas, and for duplicated collecting systems with a diseased moiety. If the partial nephrectomy is being done for renal cell carcinoma, it may be accompanied with a **regional lymphadenectomy**. The flank approach ([Fig. 9.10A](#)) is well suited for this operation and begins with an incision over the 12th or 11th rib, or in between, and extends anteriorly over the external and internal oblique muscles, which are transected. The transversalis muscle and fascia are opened, exposing Gerota fascia. The renal capsule is exposed at the planned site of resection. Control of the renal vessels and local hypothermia with ice slush is advised for control of bleeding, if

excessive. Incision in the renal parenchyma is made by sharp and blunt dissection, suture-ligating all bleeders. If the collecting system is opened, it should be closed with absorbable sutures. After complete hemostasis, Gelfoam, thrombin, Floseal, or perinephric fat is used to cover the raw surface of the kidney. More recently, robotic-assisted partial nephrectomy has been shown to provide at least equivalent outcomes compared to laparoscopic or open surgery.

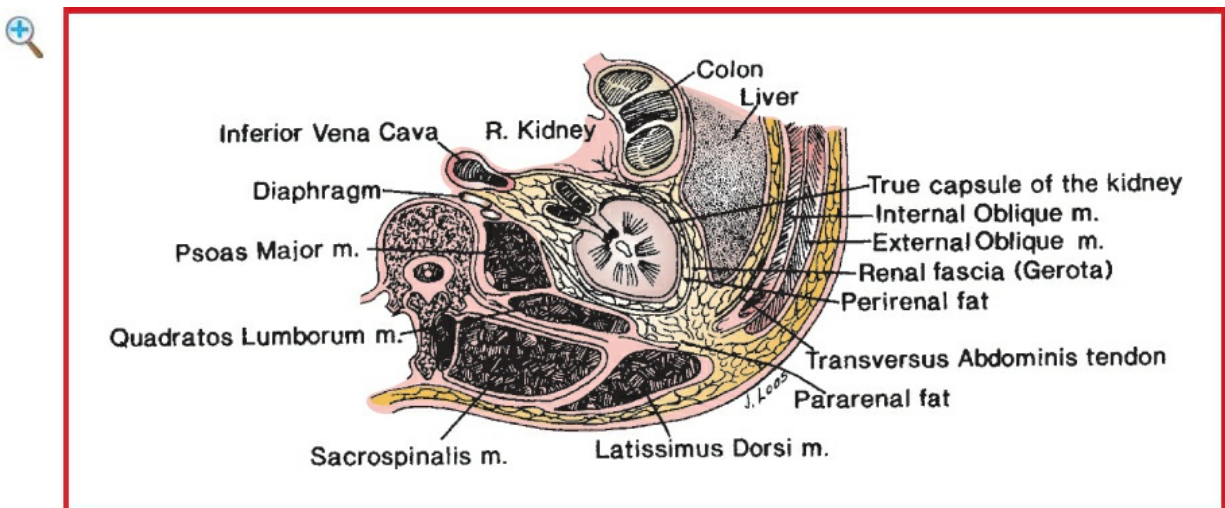


Figure 9.9 Transverse section showing the relation of the renal fascias to the right kidney. (Reproduced with permission from Baker RJ, Fischer JEMastery of Surgery, 4th edition. Lippincott Williams & Wilkins: 2001.)

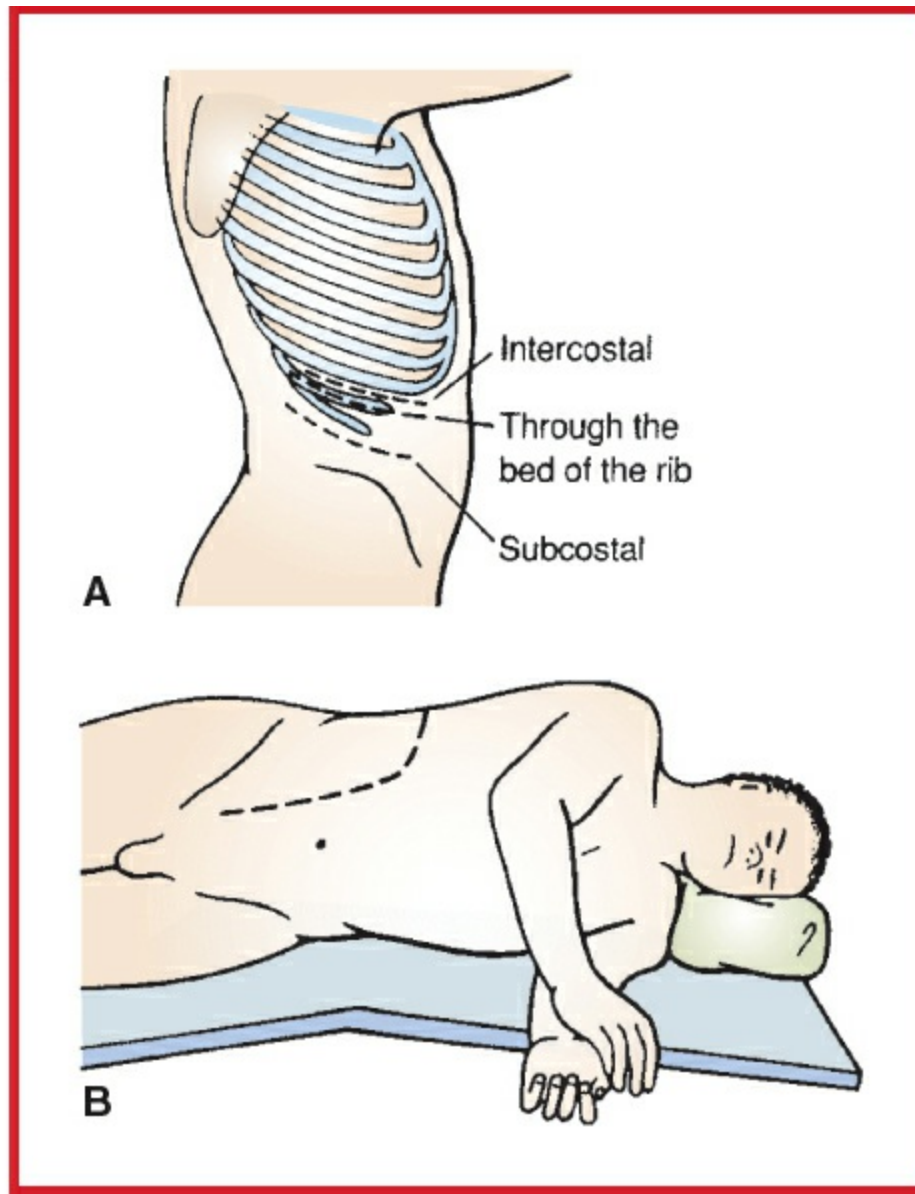


Figure 9.10 A: Flank incisions. B: Subcostal transabdominal incision.

Usual preop diagnosis

Renal cell carcinoma; double collecting system

Radical nephrectomy is the surgical excision of the kidney with its surrounding perinephric fat and Gerota fascia and the proximal 2/3 of the ureter, accompanied by paracaval or paraaortic **lymphadenectomy**. It is performed for renal cell carcinoma. Early control of renal vessels is advised before excessive manipulation of the tumor to minimize blood loss and hematogenous spread. Transabdominal or flank approaches (Fig. 9.10A, B) are best suited for this operation. Surgery in patients with renal vein or inferior vena cava involvement is more complex and prone to more intraoperative complications, including blood loss. When the tumor thrombus involves a large segment

of the vena cava or is in the right atrium, a team approach with cardiac surgeons is used. Often patients have to be put on cardiopulmonary bypass under hypothermia.

Laparoscopic simple or radical nephrectomy: A pneumoperitoneum is created by insufflating CO₂ to a pressure of 14–16 mm Hg. Three to four trocars are inserted as necessary (Fig. 9.11). Typically, the patient is placed in a flank position as in an open radical nephrectomy. This procedure can be either **transperitoneal** or **retroperitoneal**, depending on the surgeon's preference or experience. The transperitoneal approach may be modified by using a hand-assistance device through one of the ports (**hand-assisted laparoscopic nephrectomy**). Most radical nephrectomies can be done by an experienced laparoscopic surgeon.

Usual preop diagnosis

Renal cell carcinoma, nonfunctioning kidney 2° infection or obstruction; kidney donation for transplant

Nephroureterectomy is a radical nephrectomy with ureter resection, including the ureteral orifice and a cuff of bladder wall around it. It is accompanied by a regional lymphadenectomy because it is performed for a cancerous condition. The approach is either **transabdominal** or **extraperitoneal** through an extended flank incision, starting at the tip of the 11th rib and curving caudally along the lateral edge of the rectus abdominis muscle down to the pubic bone (Fig. 9.10B). Some surgeons prefer two separate incisions: a flank incision for the radical nephrectomy part and a lower abdominal incision for the ureterectomy. This procedure can be done by laparoscopy with a lower midline incision made to remove the specimen. It is often done as a combined laparoscopic and open procedure. The kidney is removed laparoscopically, while the distal ureter is excised by open surgery. The specimen is removed through the infraumbilical incision. This technique involves positioning, prepping, and draping the patient twice during the case.

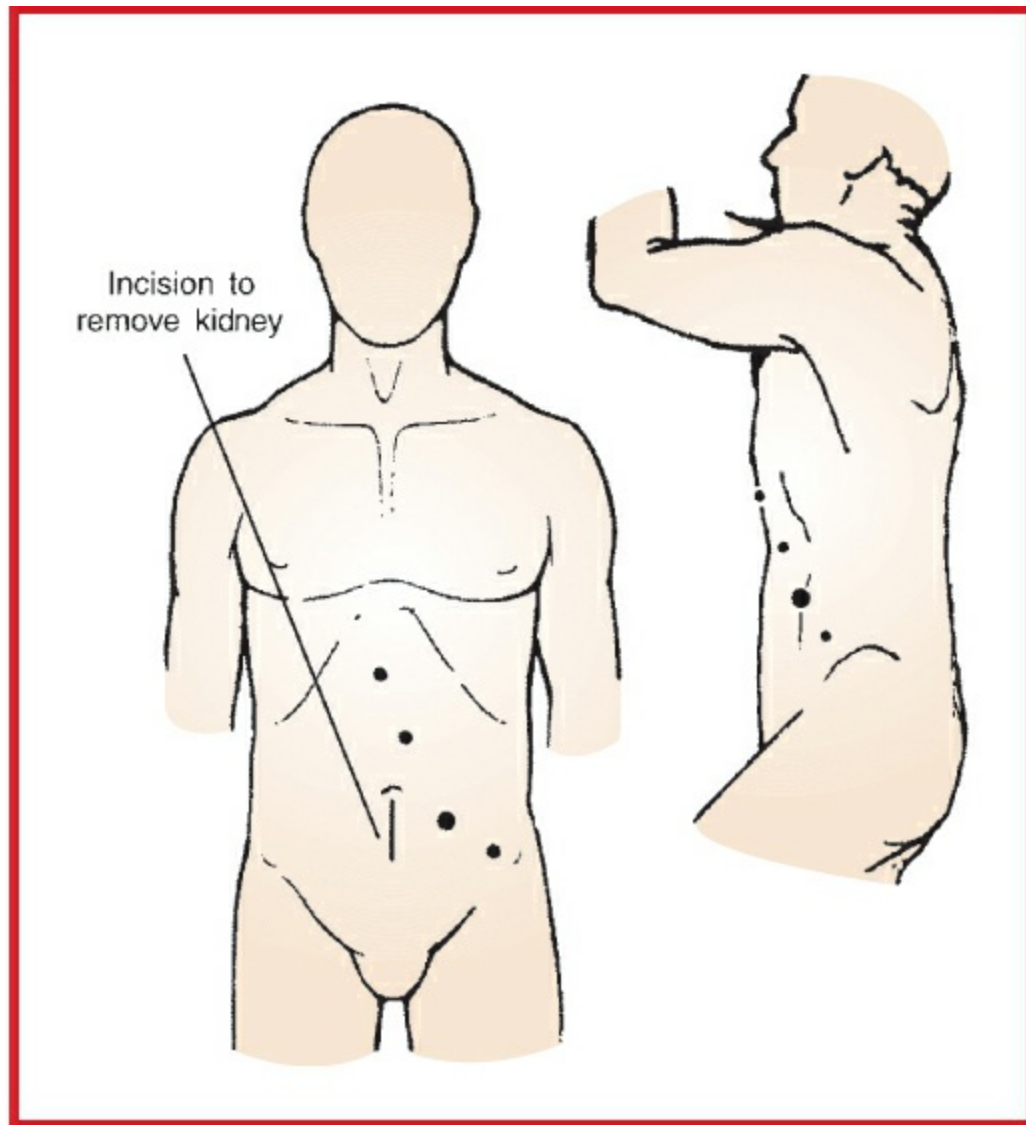


Figure 9.11 Incision and placement of trocars in laparoscopic nephrectomy. (Reproduced with permission from Scott-Conner CEH, Dawson DLOperative Anatomy, 2nd edition. Lippincott Williams & Wilkins: 2003.)

Usual preop diagnosis

Transitional cell carcinoma of the renal collecting system or ureter

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, postop pain management options, postop recovery plan, possibility of NG tube, and urinary drain postop. See [Appendix K, p. K-1](#).

SUMMARY OF PROCEDURES

	Simple Nephrectomy	Partial Nephrectomy	Radical Nephrectomy	Laparoscopic Nephrectomy or Nephroureterectomy
Position	Flank or prone	Flank	Supine or flank	Flank
Incision	Flank (Fig. 9.10A) or dorsal along paraspinous muscles	Flank (Fig. 9.10A)	Midline or subcostal trans-abdominal (Fig. 9.10B) or flank; subcostal or intercostal or through bed of 11th or 12th rib (Fig. 9.10A)	3–4 ports (Fig. 9.11)
Unique considerations	None	⇐	If tumor involves renal vein and/or IVC, clamp IVC.	Risk of emergent conversion to open case
Antibiotics	Cefazolin 2 g iv	⇐	⇐	⇐
Surgical time	2–3 h	3–4 h	⇐	⇐
Closing considerations	Chest tube may be required if pleura opened with flank incision.	⇐	⇐	A small incision is made at the end of the case to remove the specimen.
EBL	500 mL	1200 mL	500 mL	300 mL
Mortality	<1%	⇐	1%	⇐
Morbidity	Prolonged ileus: 5% Pneumothorax 2° unrecognized pleural perforation: 2%	⇐	⇐	Vascular injury requiring conversion to open case
Pain score	10	10	10	4

PATIENT POPULATION CHARACTERISTICS

Age range	All ages	⇐	⇐	⇐
Male:Female	1:1	⇐	⇐	⇐
Incidence	<1%	⇐	⇐	⇐
Etiology	Double collecting system; chronic hydronephrosis; hypoplastic kidney; renovascular HTN	Localized renal cell carcinoma	Wilms tumor (8% of all childhood malignancies); transitional cell carcinoma (7% of all kidney tumors); renal cell carcinoma (3% of adult malignancies)	
Associated conditions	HTN (if nephrectomy was used to treat renovascular HTN)			

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations following Operations on the Renal Pelvis and Upper Ureter.

Suggested Readings

1. Cacciamani GE, Medina LG, Gill T, et al: Impact of surgical factors on robotic partial nephrectomy outcomes: comprehensive systematic review and meta-analysis. J Urol 2018; 200(2):258–74.
2. Conacher ID, Soomro NA, Rix D: Anaesthesia for laparoscopic urological surgery Br J Anaesth 2004; 93(6):859–64.
3. Piros L, Langer RM: Laparoscopic donor nephrectomy techniques. Curr Opin Organ Transplant 2012; 17(4):401–5.

OPERATIONS ON THE RENAL PELVIS AND UPPER URETER

SURGICAL CONSIDERATIONS

Description

Operations on the renal pelvis and upper ureter are becoming less common because of the increasing use of endoscopic and percutaneous procedures. The basic surgical approach is the same as that for nephrectomy. Specific procedures include the following:

Pyeloplasty is the surgical correction of congenital ureteropelvic junction stenosis to relieve obstruction. The most commonly used is the **dismembered pyeloplasty**, or **Anderson-Heinz pyeloplasty**, wherein the diseased ureteropelvic junction is excised, the redundant renal pelvis is reduced, and an anastomosis is established between the renal pelvis and ureter ([Fig. 9.12](#)).

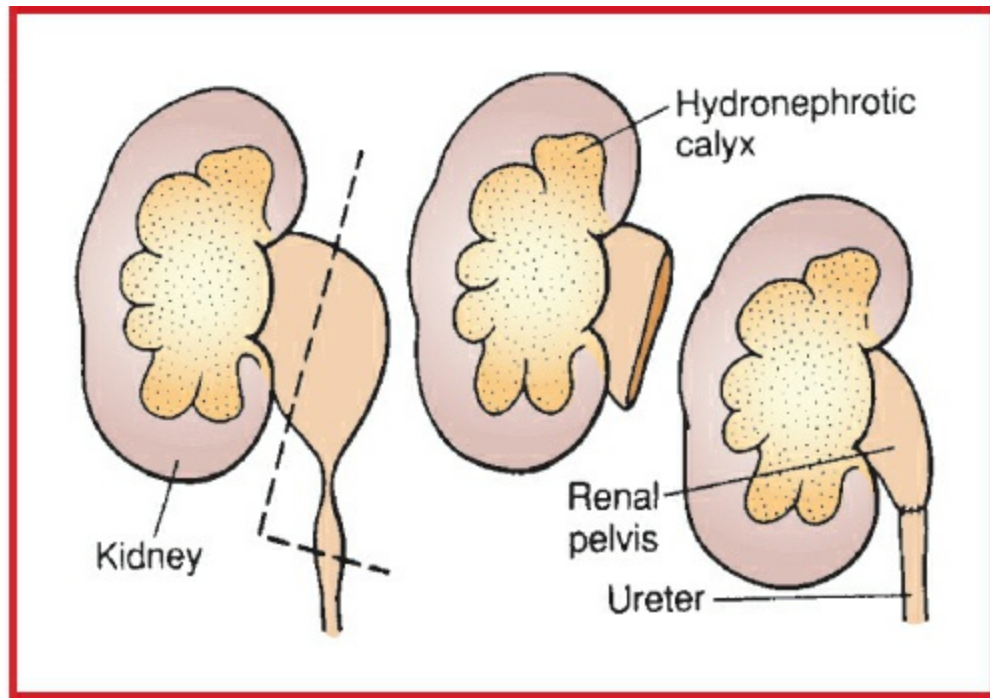


Figure 9.12 Dismembered pyeloplasty.

Usual preop diagnosis

Ureteropelvic junction obstruction

Pyelolithotomy and ureterolithotomy are used to remove calculi from the renal pelvis or ureter. The upper ureter and renal pelvis are exposed, usually through a flank approach, the calculus is palpated, and an incision is made in the ureter or renal pelvis over the calculus, which is then delivered. The incision is closed with fine, absorbable sutures.

Usual preop diagnosis

Renal pelvic or ureteral stone

Transureteroureterostomy is the transposition of one ureter across the midline and anastomosing it to the other ureter (Fig. 9.13). This operation is performed whenever the distal ureter is traumatized or diseased, and the proximal ureter is not long enough to reimplant into the bladder. The recipient ureter should be normal.

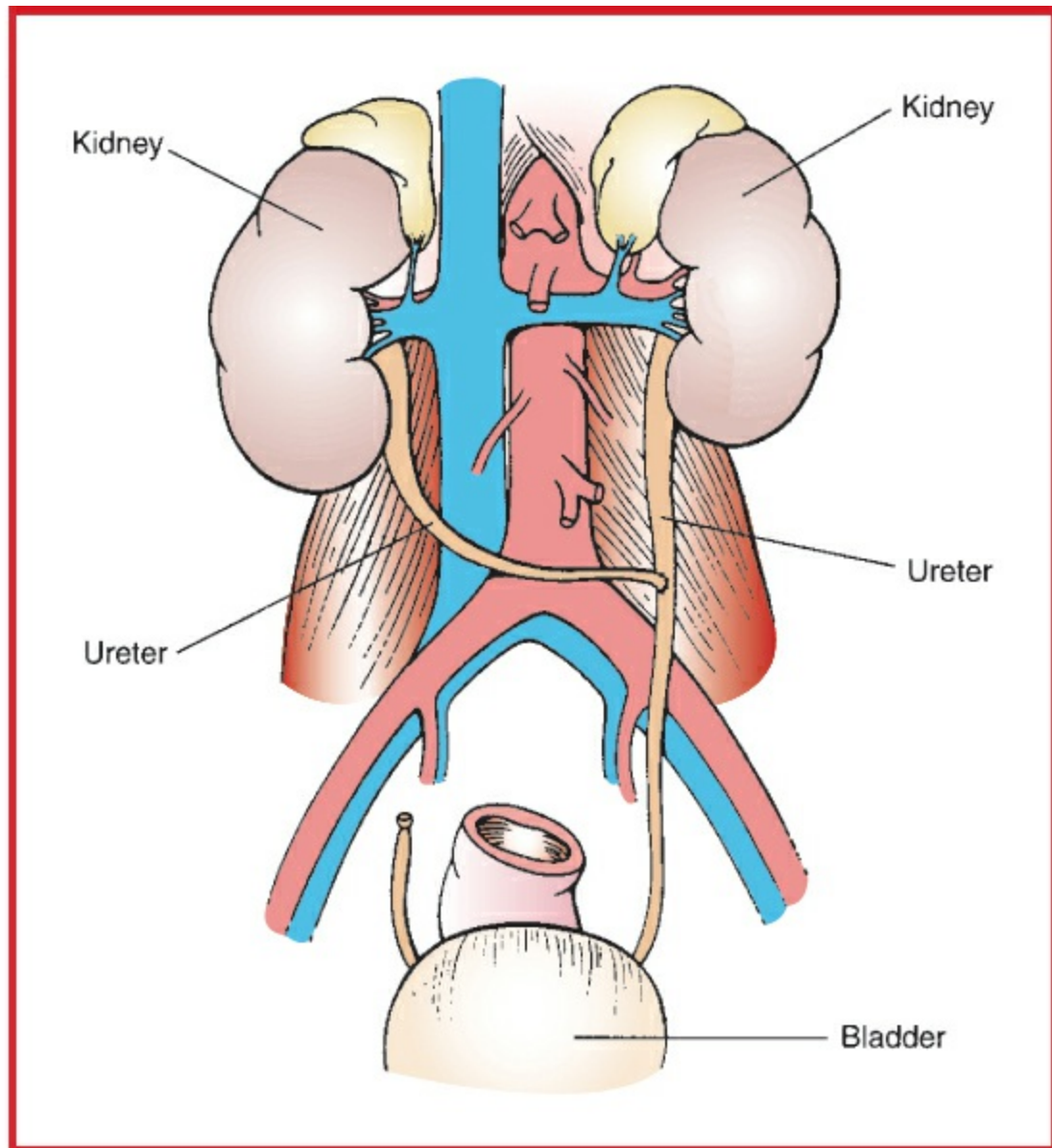


Figure 9.13 Transureteroureterostomy.

Usual preop diagnosis

Traumatic loss of distal ureter; distal ureteral tumor requiring distal ureterectomy

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, postop pain management options, postop recovery plan, possibility of NG tube, and urinary drain postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES

	Pyeloplasty	Pyelolithotomy/Ureterolithotomy	Transureteroureterostomy
Position	Flank or prone	⇐	Supine
Incision	Flank (Fig. 9.10A) or dorsal	⇐	Midline abdominal (Fig. 9.10A, inset)
Antibiotics	Cefazolin 2 g iv	⇐	⇐
Surgical time	3 h	1–2 h	3 h
EBL	Minimal	⇐	⇐
Mortality	<1%	⇐	⇐
Morbidity	Urinary leakage: 5% Infection: 2%	⇐ ⇐	⇐ ⇐
Pain score	10	10	10

■ PATIENT POPULATION CHARACTERISTICS

Age range	All ages	⇐	⇐
Male:Female	1:1	⇐	⇐
Incidence	Rare	Extremely rare	⇐
Etiology	Ureteropelvic junction obstruction: 80% of causes of dilated collecting system in the newborn	Renal pelvic and upper ureteral stone	Traumatic loss of lower ureter; lower ureteral tumor
Associated conditions	Renal failure: 1%		

ANESTHETIC CONSIDERATIONS

(Procedures covered: nephrectomy and operations on the renal pelvis and upper ureter.)

PREOPERATIVE

Patients presenting for nephrectomy and operations on the renal pelvis and upper ureter may be of any age, depending on the etiology of the abnormality. Many patients may have renal insufficiency 2° the underlying problem or from renovascular HTN. Elderly patients frequently have preexisting medical problems, including CAD, CHF, PVD cerebrovascular disease, COPD, and renal impairment. Preop evaluation should be directed toward the detection and treatment of these conditions prior to anesthesia.

Respiratory	Increased postop pulmonary complications because of location of incision (nonlaparoscopic). If Hx of pulmonary disease (e.g., asthma, COPD), consider postop respiratory therapy. Tests: As indicated from H&P
Cardiovascular	Consider possibility of renal HTN.
Hematologic	Polycythemia may be seen in association with polycystic kidney disease, renal cell carcinoma. Consider preop blood donation for autologous transfusion. Tests: Hct
Laboratory	Electrolytes; BUN; Cr; other tests as indicated from H&P
Premedication	Standard premedication
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of a rectus sheath catheter or thoracic epidural block for postop pain management. Consider preop acetaminophen 1000 mg, ± celecoxib 200 mg, and ±gabapentin 300 mg po with sip of water. Consider aprepitant 40 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA is recommended for these procedures; technique depends on underlying disease. Regional techniques (spinal or epidural) may be alternatives for some open procedures but are less than optimal because of awkward positioning that may → patient discomfort and pain, resulting from diaphragmatic stimulation. Consider combined technique with regional opiates.

Induction	Standard induction	
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider preincision placement of nerve blocks.	
Maintenance	Standard maintenance. If intraperitoneal or laparoscopic approach is used, surgeon may request limiting N ₂ O to avoid distension of bowel and interference with operative field.	
Emergence	No specific considerations. Discuss with surgeon local wound infiltration or nerve block placement or reinforcement if initially placed >4 h ago. Consider iv acetaminophen if po dose is given >6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.	
Blood and fluid requirements	Mild-to-moderate blood loss IV: 14–16 ga × 1 BSS @ 2–5 mL/kg/h Warm all fluids.	Intraperitoneal approach associated with higher vasopressor use and fluid requirements. When renal vessels are to be cross-clamped, mannitol (0.5 g/kg) is often given prior to occlusion (20 min maximum). See Goal-Directed Fluid Management, Appendix K , p. K-4.

Monitoring	Standard monitors Urinary catheter Arterial line (partial nephrectomy) ± CVP line	Invasive monitoring if indicated from H&P. CVP line may be useful for partial nephrectomy in patients with solitary kidneys.
Positioning	Use axillary roll if lateral. Avoid stretching brachial plexus—limit abduction to 90° If prone, repeatedly ✓ eyes and pressure points. Assure free excursion of the abdomen	The lateral position with kidney rest and table flexion may → ↓ BP, possibly 2° vena cava obstruction. Vasopressors and moderate iv volume administration with gradual assumption of the position are recommended to avoid this complication.
Complications	Pneumothorax ↓ BP with positioning (see above). Indigo carmine → ↑ BP and ↑ SVR Methylene blue → ↓ BP	Sx of pneumothorax include ↑ RR, ↑ PIP, hypoxemia, and hypercarbia. If in doubt, ✓ CXR.

POSTOPERATIVE

Complications	Postnephrectomy syndrome Eye injury (if prone) Brachial plexus injury (if lateral) Pneumothorax Atelectasis Pneumonia	Postnephrectomy syndrome 2° retractor injury: L1 nerve root damage with resulting pain, dyesthesia, and sensory loss in L1 dermatome distribution

Multimodal pain management	Hydromorphone Acetaminophen Celecoxib PCA Consider epidural opiate.	Postop analgesia critical to minimize pulmonary complications. Emphasize multimodal approach combining nonopioid and bolus/PCA opioid analgesics with epidural analgesia, rectus sheath, and/or TAP blocks (see Appendix K, p. K-6).
ERAS	PONV protocol (see Appendix B, p. B-4 and p. B-6)	Encourage early enteral nutrition, Foley catheter removal, and mobilization. Consider DVT prophylaxis. Chewing gum promotes return of bowel function.
Tests	Hct CXR	Others dependent on operative course, coexisting disease

Suggested Reading

1. Vukovic N, Dinic L: Enhanced recovery after surgery protocols in major urologic surgery. *Front Med (Lausanne)* 2018; 5:93.

CYSTECTOMY

SURGICAL CONSIDERATIONS

Description

Open (in contrast to transurethral or endoscopic) bladder operations (cystectomies) account for 15–20% of all urologic procedures. They are grouped as simple, partial, and radical procedures.

Simple cystectomy is performed for benign conditions of the bladder, such as severe hemorrhagic cystitis, radiation cystitis, and contracted bladder. It involves the removal of the bladder only. The operation is performed through a lower abdominal incision.

The peritoneal reflections are incised down to the pouch of Douglas; the vasa deferentia and superior vesical arteries are identified, cross-clamped, transected, and tied. The ureters are identified, separated from the surrounding tissues, cross-clamped near the bladder, transected, and tied. The bladder is bluntly separated from the anterior rectal wall all the way to the apex of the prostate. The lateral pedicles of the bladder are cross-clamped, cut, and tied. The endopelvic fascia is incised, separating the prostate from the lateral pelvic wall. The puboprostatic ligaments are transected, and the dorsal vein of the penis is suture-ligated. The tied dorsal vein and urethra are incised just distal to the apex of the prostate. The specimen is delivered out of the incision and hemostasis secured with electrocautery. An ileal conduit is then performed (see below).

Partial cystectomy is the excision of only the part of the bladder containing the pathology. This is not a commonly performed operation and is reserved for tumors located in the dome of the bladder of older patients who are poor surgical risks for major operations, such as radical cystectomy. The operation is preceded by a cystoscopy to identify the site of pathology. Beginning with a lower abdominal incision, the dome and lateral walls of the bladder are separated from the surrounding tissues, which are covered by wet packs to minimize contamination. An incision is made in the dome of the bladder at least 2 cm away from the pathology. The inside of the bladder is inspected and the pathology identified. The incision in the bladder is continued around, and at least 2 cm away from, the pathology, until the latter is completely excised. Bleeders in the wall of the bladder are electrocoagulated. The bladder wall is then closed in two layers—a through-and-through layer and an inverting layer—using absorbable material. Wet packs are removed, a drain is left in the region, and the abdominal incision is closed.

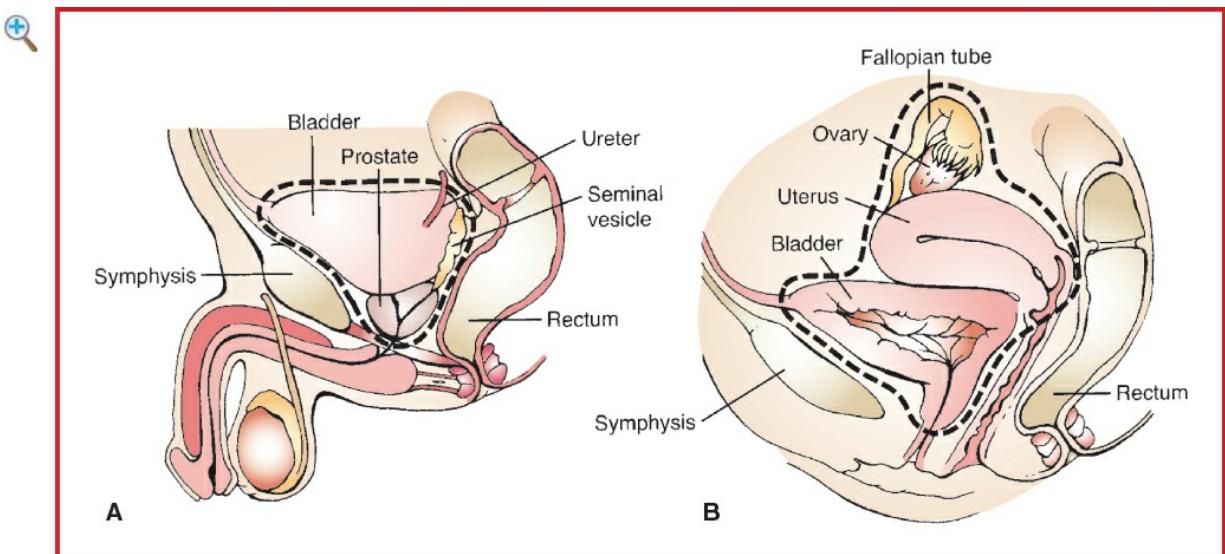


Figure 9.14 Anatomy of the pelvis with tissue to be excised outlined by dashed line. **A:** Male. **B:**

Female.

Radical cystectomy (or **radical cystoprostatectomy**) is performed for treatment of invasive bladder cancer. It encompasses the removal of the bladder and the lower ureters, the prostate gland, and seminal vesicles in men ([Fig. 9.14A](#)) and the uterus, ovaries, and the anterior vaginal wall in women ([Fig. 9.14B](#)). Accompanied by a **pelvic lymphadenectomy**, it is performed in the supine position, except when a concomitant **urethrectomy** is required, wherein a lithotomy position is used.

Following cystectomy, whether radical or simple, some form of **urinary diversion** is required. This can be accomplished with either a standard ileal conduit or a bladder substitution. The **ileal conduit** is constructed from 6 to 8 inches of the terminal ileum isolated, with its blood supply, from the small intestine. The continuity of the small intestine is accomplished by a simple anastomosis. The ureters are implanted into the proximal end of the conduit, and the distal end is brought through the abdominal wall as a stoma ([Fig. 9.15](#)). **Bladder substitution** is a more complex operation wherein a longer segment of bowel is isolated, with its blood supply, and fashioned into a pouch. The ureters are implanted in the pouch, and the most dependent part of the pouch is connected to the membranous urethra, avoiding a stoma ([Fig. 9.16](#)). Not all patients undergoing cystectomies are candidates for bladder substitution. For example, patients who require a urethrectomy are not candidates because of the need to remove the urethra.

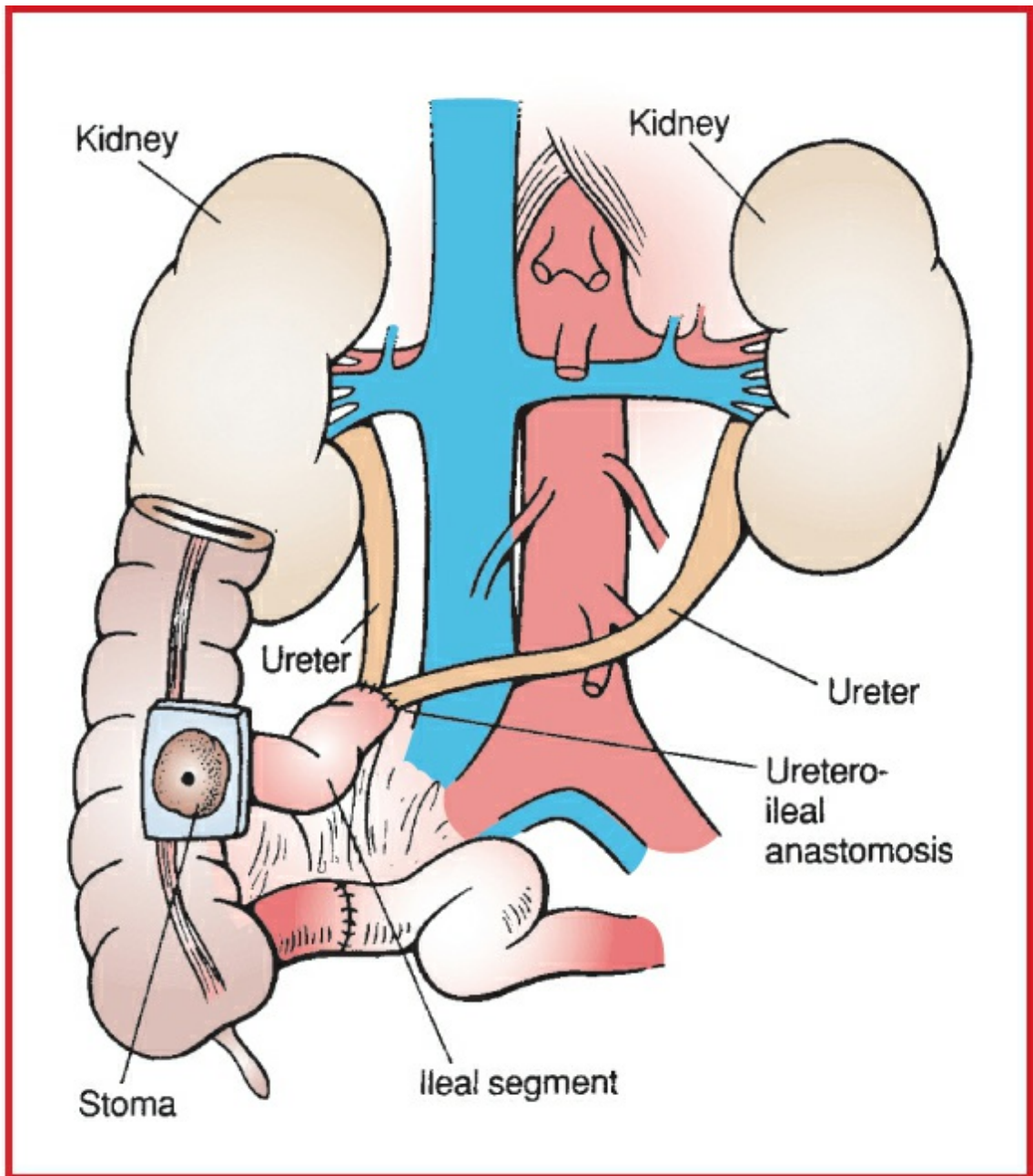


Figure 9.15 Ileal conduit: A segment of the ileum is isolated from the terminal ileum, and continuity of the bowel is reestablished with an end-to-end anastomosis. Ureters are joined to the proximal end of the ileal segment, and the distal end is brought out to the skin as a stoma.

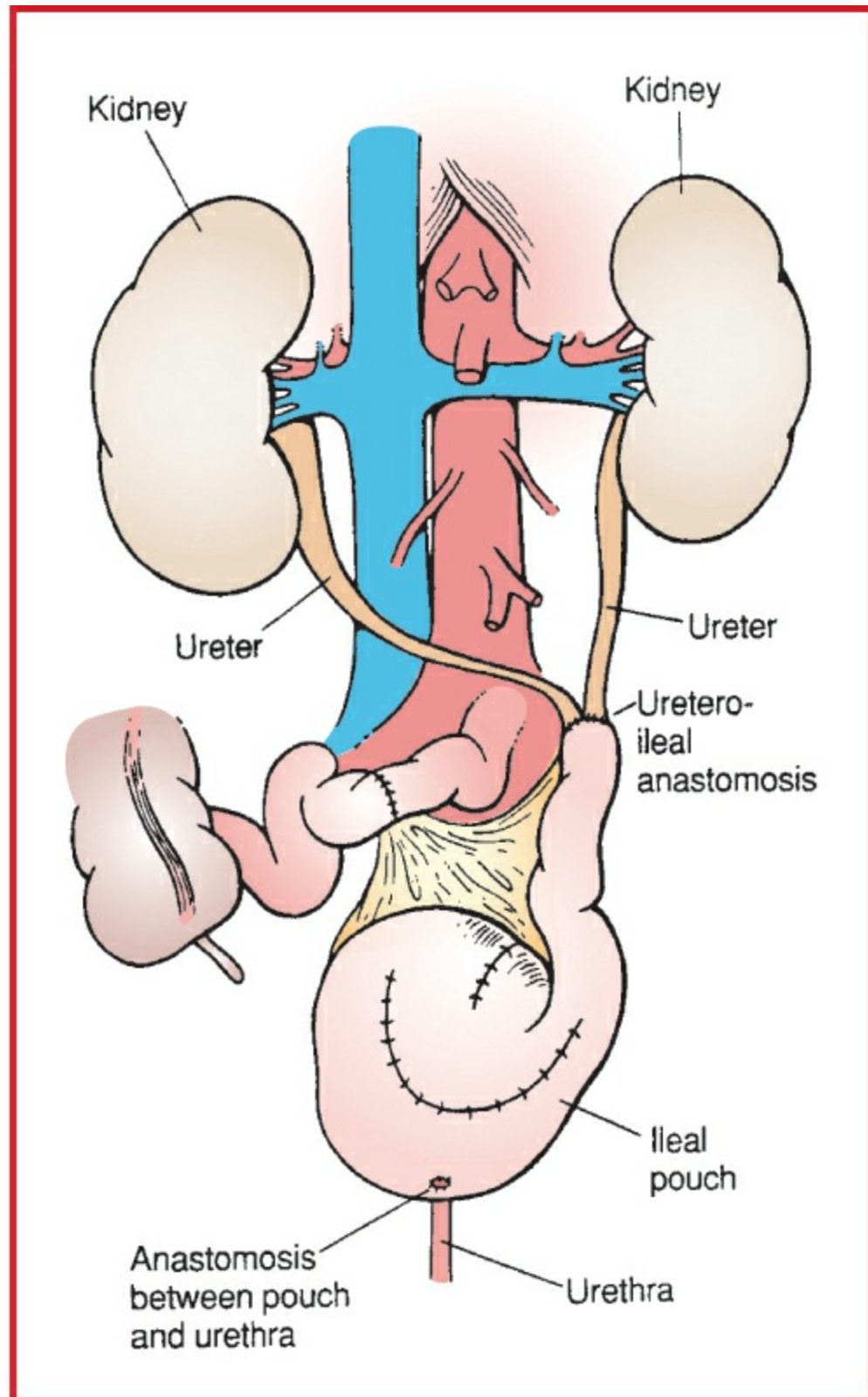


Figure 9.16 Bladder substitution: A segment of the ileum is fashioned into a pouch and anastomosed to the urethra. The ureters are joined to the proximal, nondetubularized segment.

Usual preop diagnosis

Bladder cancer; contracted bladder; hemorrhagic cystitis; radiation cystitis; bladder

diverticulum

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, DVT prophylaxis, postop pain management options, postop recovery plan, possibility of NG tube, and urinary drain postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES			
	Simple Cystectomy	Partial Cystectomy	Radical Cystectomy
Position	Supine	⇐	Supine or lithotomy
Incision	Transperitoneal, midline	⇐	⇐
Antibiotics	Cefotetan or ceftriaxone 1 g	⇐	⇐
Surgical time	4 h	2 h	6 h
EBL	1000 mL	Minimal	1500 mL
Postop care	Care of the stoma	Catheter care	Care of the stoma
Mortality	1%	<1%	2%
Morbidity	Prolonged ileus: 5% Infection: 2%	— — Hematuria: 5%	5% 2%
Pain score	10	10	10

■ PATIENT POPULATION CHARACTERISTICS			
	Simple Cystectomy	Partial Cystectomy	Radical Cystectomy
Age range	40–80 yr	⇐	⇐
	3:1	⇐	⇐

Male:Female			
Incidence	60,000 new cases of bladder cancer diagnosed/yr; 20% treated with cystectomy	⇐	⇐
Etiology	Contracted bladder; hemorrhagic and radiation cystitis	Bladder cancer; bladder diverticulum	⇐
Associated conditions	Heart disease (10%); HTN (10%); COPD (5%); diabetes mellitus (5%)	Heart disease (10%)	⇐

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Patients presenting for cystectomy are frequently elderly and may have preexisting medical problems, including CAD, CHF, PVD, cerebrovascular disease, COPD, and renal impairment. Preop evaluation should be directed toward the detection and treatment of these conditions prior to anesthesia.

Respiratory	✓ for pulmonary disease in older patients Test: As indicated from H&P
Cardiovascular	✓ for cardiac disease, HTN in older patients Tests: Consider ECG; others as indicated from H&P
Gastrointestinal	Bowel prep is likely and may cause dehydration and electrolyte disturbances. Tests: Electrolytes, if indicated
Hematologic	T&C for 2–4 U PRBC Test: Hct
Laboratory	Other tests as indicated from H&P
Premedication	Sedation prn anxiety in adults (e.g., lorazepam 1–2 mg po 1–2 h preop; midazolam 1–2 mg iv in preop)

	area)
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of a rectus sheath catheter or thoracic epidural block for postop pain management. Consider preop acetaminophen 1000 mg, ± celecoxib 200 mg, and ±gabapentin 300 mg po with sip of water. Consider aprepitant 40 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

Spinal, continuous lumbar epidural, or GA is acceptable, with choice dependent on the length of procedure, coexisting disease, and patient preference. A combined technique using GA and regional anesthesia may be preferable. A T4 sensory level is recommended, because peritoneal stimulation is likely during this procedure.

Regional anesthesia

Spinal	0.75% bupivacaine 12–15 mg in 7.5% dextrose; hyperbaric tetracaine 10–15 mg with 200 mcg epinephrine for procedures >3 h
Epidural	1.5–2% lidocaine with epinephrine 5 mcg/mL, 15–25 mL; supplement with 5–10 mL as needed. Supplemental iv sedation (e.g., midazolam 1 mg iv prn; fentanyl 50 mcg iv prn). Bupivacaine (0.25–0.5%), levobupivacaine (0.25–0.5%), or ropivacaine (0.5–0.75%), with or without sufentanil or fentanyl, may also be used. The addition of opiates has been associated with ↑ urinary retention in noncatheterized patients.

General anesthesia

Induction	Standard induction

ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider preincision placement of nerve blocks.	
Maintenance	Standard maintenance. Consider ketamine 0.5 mg/kg bolus + 0.25 mg/kg/h infusion during surgery iv (especially in opioid-dependent patients).	
Emergence	No specific considerations. Discuss with surgeon local wound infiltration or nerve block placement or reinforcement if initially placed >4 h ago. Consider iv acetaminophen if po dose is given >6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.	
Blood and fluid requirements	Significant blood loss possible IV: 16 ga × 1 BSS @ 6–10 mL/kg/h Warm fluids Humidify gases	Blood loss may be less with regional than GA. See Goal-Directed Fluid Management, Appendix K , p. K-4.
Monitoring	Standard monitors Arterial line CVP line	Consider PA catheter or TEE in patients with cardiopulmonary disease. UO as measure of volume status may be unavailable during procedure.
Complications	Major blood loss 3rd-space losses Hypothermia	
ERAS	PONV protocol (see Appendix B , p. B-4 and p. B-6)	Encourage early enteral nutrition, Foley catheter removal, and mobilization. Consider DVT prophylaxis. Chewing gum promotes return of bowel function.

▼ POSTOPERATIVE

Complications	Hypothermia	
Multimodal pain management	Emphasize multimodal approach combining nonopioid and bolus/PCA opioid analgesics with epidural analgesia, rectus sheath, and/or TAP blocks (see Appendix K, p. K-6).	Regional analgesia. Acetaminophen, fentanyl 100–250 mcg, or Dilaudid 0.5–1 mg, ± celecoxib 200 mg Pain is considerably less after robotic cases, and pain management should be adjusted accordingly.
ERAS	PONV protocol (see Appendix B, p. B-4 and p. B-6)	Encourage early enteral nutrition, Foley catheter removal, and mobilization. Consider DVT prophylaxis. Chewing gum promotes return of bowel function.
Test	Hct	Others as indicated by intraop course

Suggested Readings

1. Azzouni F: Current status of robot-assisted radical cystectomy for bladder cancer. *Net Rev Urol* 2012; 9(10):573–82.
2. Cheung G, Dasgupta P, Guru KA, et al: Minimally invasive cystectomy approaches in the treatment of bladder cancer. *Expert Rev Anticancer Ther* 2012; 12(6):733–41.
3. El-Galley R, Hammontree L, Urban D, et al: Anesthesia for laparoscopic donor nephrectomy: is nitrous oxide contraindicated? *J Urol* 2007; 178(1):225–7.
4. Poch MA, Raza J, Nyquist J, et al: Tips and tricks to robot-assisted radical cystectomy and intracorporeal diversion. *Curr Opin Urol* 2013; 23(1):65–71.
5. Vukovic N, Dinic L: Enhanced recovery after surgery protocols in major urologic surgery. *Front Med (Lausanne)* 2018; 5:93.

OPEN BLADDER OPERATIONS (OTHER THAN CYSTECTOMY)

SURGICAL CONSIDERATIONS

Description

Open bladder operations include the following:

Augmentation cystoplasty (or **enterocystoplasty**): Small, contracted bladders can be enlarged and their size and capacity augmented with a segment of the intestine. The bladder is opened widely, anterior/posteriorly, or from side to side or with a cruciate incision. A segment of the intestine—small bowel, cecum, or colon—is isolated from the intestinal tract, detubularized, and added onto the bladder.

Variant procedure

The antrum of the stomach can also be used (**gastrocystoplasty**).

Usual preop diagnosis

Contracted bladder from chronic cystitis or neurogenic bladder

Repair of vesicovaginal or enterovesical fistulas

The communication between the vagina and bladder or bladder and bowel is identified and excised, and the edges are freshened until normal, noninflamed tissues are exposed. The openings in the bladder and in the vagina or bowel are closed, and omentum is interposed in between to promote healing and prevent recurrence. With enterovesical fistulas, often the diseased segment of the intestine is excised, and an **end-to-end anastomosis** of the intestine is performed.

Variant procedure

Transvaginal repair of vesicovaginal fistula (see Vaginal Operations).

Usual preop diagnosis

Vesicovaginal or enterovesical fistula

Ureteral reimplantation, performed to correct vesicoureteral reflux, is more commonly used in the pediatric group than in adults. In adults, it is performed mainly for lower ureteral injuries, iatrogenic or traumatic. The lower ureter is identified and dissected proximally until adequate length is obtained. The bladder is opened and a 2- to 3-cm submucosal tunnel is created in or near the trigone, and the ureter is brought into the

tunnel and fixed with sutures. If there is a large gap between the ureter and the bladder, a **psoas hitch procedure** is necessary. The bladder is mobilized and stitched to the psoas muscle to reach the ureter. In children, if the ureter is dilated, its diameter is reduced by imbrication before reimplantation. In adults, a nonrefluxing implantation is usually not necessary if the operation is being performed for ureteral injury.

Usual preop diagnosis

Vesicoureteral reflux; lower ureteral injuries

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, discussion of carb loading, DVT prophylaxis, postop pain management options, and possibility of urinary drain postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES			
	Augmentation Cystoplasty	Repair of Fistulas	Ureteral Reimplantation
Position	Supine	Supine or lithotomy	Supine
Incision	Low abdominal	⇐	⇐
Antibiotics	Gentamicin 80 mg iv, slowly	⇐	⇐
Surgical time	4 h	3 h	⇐
EBL	Minimal	⇐	⇐
Postop care	Care of catheters and stents	⇐	⇐
Mortality	<1%	⇐	⇐
Morbidity	Infections: 5%	⇐	⇐
	Urinary leakage: 1%	⇐	⇐
Pain score	10	10	10

■ PATIENT POPULATION CHARACTERISTICS			
Age range	All ages	⇐	⇐

Male:Female	1:4	⇐	⇐
Incidence	Rare	⇐	⇐
Etiology	Contracted bladders from chronic cystitis	Traumatic fistulas; vesicovaginal; regional enteritis; diverticulitis; colon cancer	Vesicoureteral reflux; injury to lower ureters

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Patients presenting for open bladder operations may be of any age, depending on the etiology of the abnormality. Elderly patients frequently have preexisting medical problems, including CAD, CHF, PVD, cerebrovascular disease, COPD, and renal impairment. Preop evaluation should be directed toward the detection and treatment of these conditions prior to anesthesia.

Respiratory	✓ for pulmonary disease in elderly patients Tests: As indicated from H&P
Cardiovascular	✓ for cardiac disease in elderly patients Tests: Consider ECG; others, if indicated from H&P.
Neurologic	Paraplegics and quadriplegics may present for operations on the bladder and urinary tract. Obtain Hx of autonomic hyperreflexia (AH). Sx are flushing, HA, nasal stuffiness, and HTN associated with voiding or noxious stimuli below the level of transection.
Laboratory	Other tests as indicated from H&P
Premedication	Midazolam 1–2 mg iv
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po. Consider aprepitant 40 mg po with sip of water for patients with h/o severe PONV.

◆ INTRAOPERATIVE

Anesthetic technique

Spinal, continuous lumbar epidural, or GA is acceptable, with choice dependent on the length of procedure, coexisting disease, and patient preference. A combined technique using light GA with regional anesthesia is also acceptable. A T10 sensory level is sufficient to provide anesthesia for procedures on the bladder, but a T4 level is recommended if the peritoneum is opened. (See Anesthetic Considerations for Transurethral Procedures [except TURP], for patients with AH.)

Regional anesthesia

Spinal	0.75% bupivacaine 10–12 mg. For shorter procedures (<1 h), consider low-dose bupivacaine (0.75%, 7.5 mg), mepivacaine (1.5%, 45 mg), or procaine (10%, 100–150 mg). Lidocaine may be used, but the incidence of transient neurologic symptoms is significant (30%).
Epidural	1.5–2% lidocaine with epinephrine 5 mcg/mL, 15–25 mL; supplement with 5–10 mL as needed. Supplemental iv sedation. Bupivacaine (0.25%), levobupivacaine (0.25–0.5%), or ropivacaine (0.5–0.75%), with or without sufentanil or fentanyl, may also be used. The addition of opiates has been associated with ↑ urinary retention in noncatheterized patients.

General anesthesia

Induction	Standard induction
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.
Maintenance	Standard maintenance. Consider limiting N ₂ O for long intraperitoneal procedures to minimize bowel distension.
Emergence	No specific considerations. Consider iv acetaminophen if po dose is given >4–6 h ago. PONV

	prophylaxis should include ondansetron.	
Blood and fluid requirements	Minimal-to-moderate blood loss IV: 16–18 ga × 1 BSS @ 2–4 mL/kg/h Warm fluids Humidify gases for lengthy procedures	See Zero Balance Fluid Management, Appendix K, p. K-2 and p. K-4. Intraperitoneal procedures have higher volume requirements. If possible, use Goal-Directed Fluid Management (see p. K-4).
Monitoring	Standard monitors for simpler procedures ± Arterial/CVP lines	UO as a measure of volume status may be unreliable and often unavailable during the procedure. Consider arterial line, CVP for longer, more complex procedures.
Positioning	✓ and pad pressure points ✓ eyes	NB: In lithotomy position, peroneal nerve compression at the lateral fibular head → foot drop
Complications	AH in spinal cord–injured patients	See discussion in Anesthetic Considerations for Transurethral Procedures.

POSTOPERATIVE

Complications	Hypothermia Fever, bacteremia	
Multimodal pain	Consider epidural	Emphasize multimodal

management	narcotics or PCA.	approach combining nonopioid (e.g., acetaminophen, celecoxib) and bolus opioid analgesics (see Appendix K, p. K-6).
ERAS	PONV protocol (see Appendix B, p. B-4 and p. B-6)	Encourage early enteral nutrition, Foley catheter removal, and mobilization. Consider DVT prophylaxis. Chewing gum promotes return of bowel function.
Tests	Hct Blood cultures if febrile	Others as indicated by intraop course

Suggested Reading

1. Vukovic N, Dinic L: Enhanced recovery after surgery protocols in major urologic surgery. *Front Med (Lausanne)* 2018; 5:93.

INGUINAL OPERATIONS

SURGICAL CONSIDERATIONS

Description

Inguinal operations are very common and are usually performed on an outpatient basis. Groin dissection, however, may necessitate inpatient care.

Inguinal herniorrhaphy: A 3" inguinal incision is made, starting 1" medial to the anterior-superior iliac spine and ending at the pubic tubercle. The external oblique aponeurosis is excised, opening the external inguinal ring. The spermatic cord and the hernial sac are freed off the inguinal canal; then, the hernial sac is dissected off the spermatic cord and followed proximally into the internal inguinal ring, where it is suture-ligated and excised. The floor of the inguinal canal is strengthened by approximating the conjoined tendon to the reflected part of the inguinal ligament. Modifications include laparoscopic repair and use of surgical mesh. Please see [Section 7.9](#) for details.

Usual preop diagnosis

Inguinal hernia (See Figs. 9.17 and 9.18 for details of anatomic relationships.)

Orchiopexy is performed through the same incision as used in herniorrhaphy. After the inguinal canal is exposed, a search for the undescended testis begins. The testis and cord are dissected free from all surrounding tissue until adequate length is obtained to bring the testis down to the scrotum. Next, a pouch is created in the wall of the scrotum by incising the scrotal skin and dissecting it off the dartos fascia. The testis is brought down into the pouch and fixed to the dartos fascia with sutures, and the incisions are closed. Often a **herniorrhaphy** is performed at the same time.

Unusual preop diagnosis

Undescended testis

Radical orchiectomy is performed through a herniorrhaphy incision (described above). The spermatic cord is freed and cross-clamped at the internal inguinal ring, transected, and suture-ligated. The testis, with its tunica vaginalis, is then delivered through the incision by blunt and sharp dissection, and the inguinal incision is closed. Sometimes, a testicular prosthesis is inserted and fixed in the scrotum before the inguinal incision is closed.

Usual preop diagnosis

Testicular cancer

Ligation of spermatic vein is performed through a small, transverse incision 1–2" above the internal inguinal ring. Muscles are split and the peritoneum is reflected medially to expose the spermatic vessels; the vein is identified and ligated. This operation can also be done through a scrotal incision.

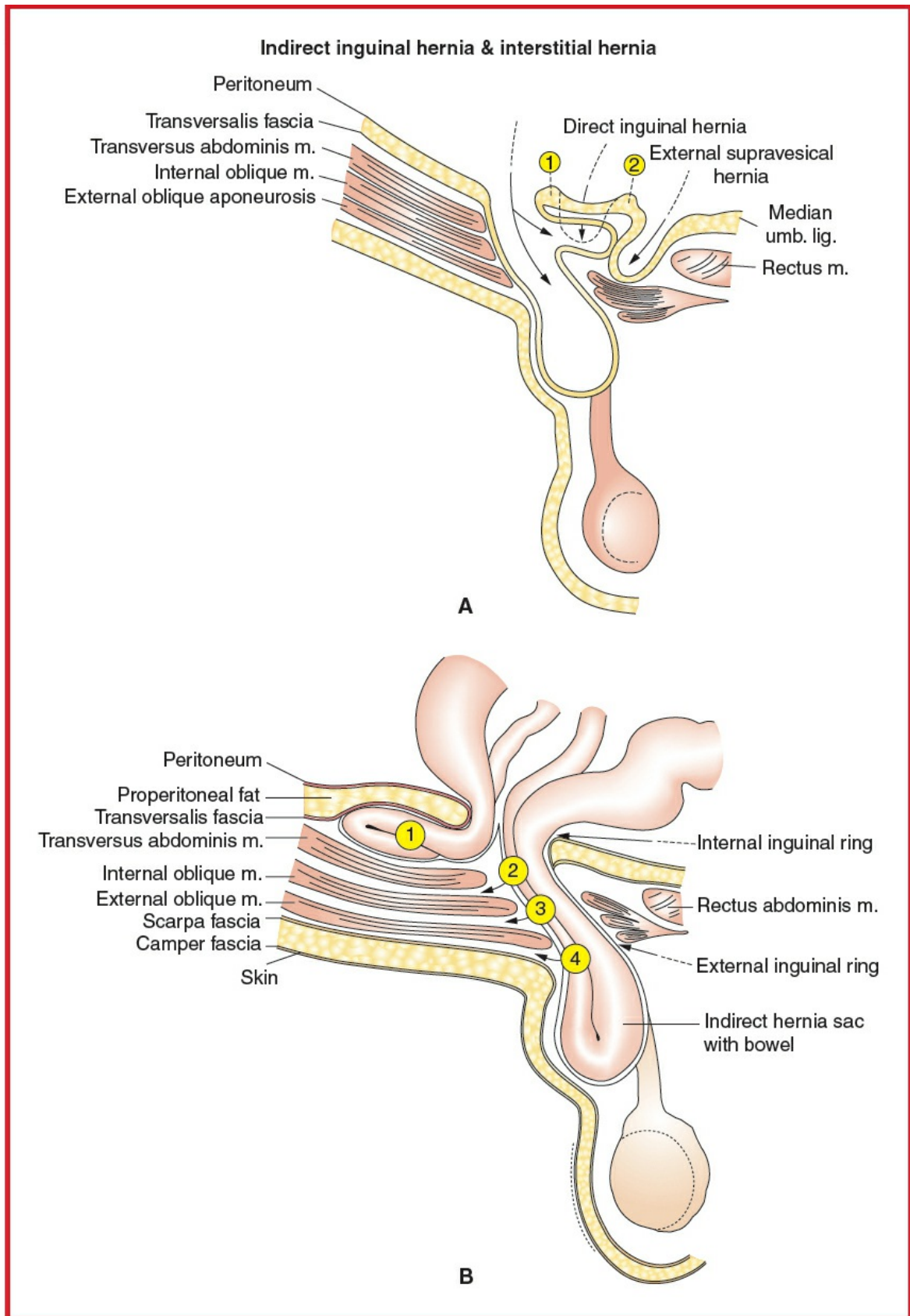


Figure 9.17 A: Relationships of four groin hernias in one patient. Indirect inguinal hernia with intraparietal diverticulum, direct hernia, and external supravesical hernia. 1, lateral umbilical ligament; 2, medial umbilical ligament. Arrows indicate different sites of origin of each hernia. (Reproduced with permission from Skandalakis JE, Gray SW, Burns WB, et al: Internal and external supravesical hernia.

Am Surg 1976; 42(2):142–6.) **B:** Diagram of intraparietal hernia. The sac, entering at the internal ring, may pass into any one or more spaces between layers of the abdominal wall. 1, properitoneal; 2 and 3, interstitial; 4, superficial. An indirect hernia also may be present. (Reprinted from Skandalakis JE, Gray SW, Akin JT Jr: The surgical anatomy of hernial rings. Surg Clin North Am 1974; 54(6):1227–46. Copyright © 1974 Elsevier. With permission.)

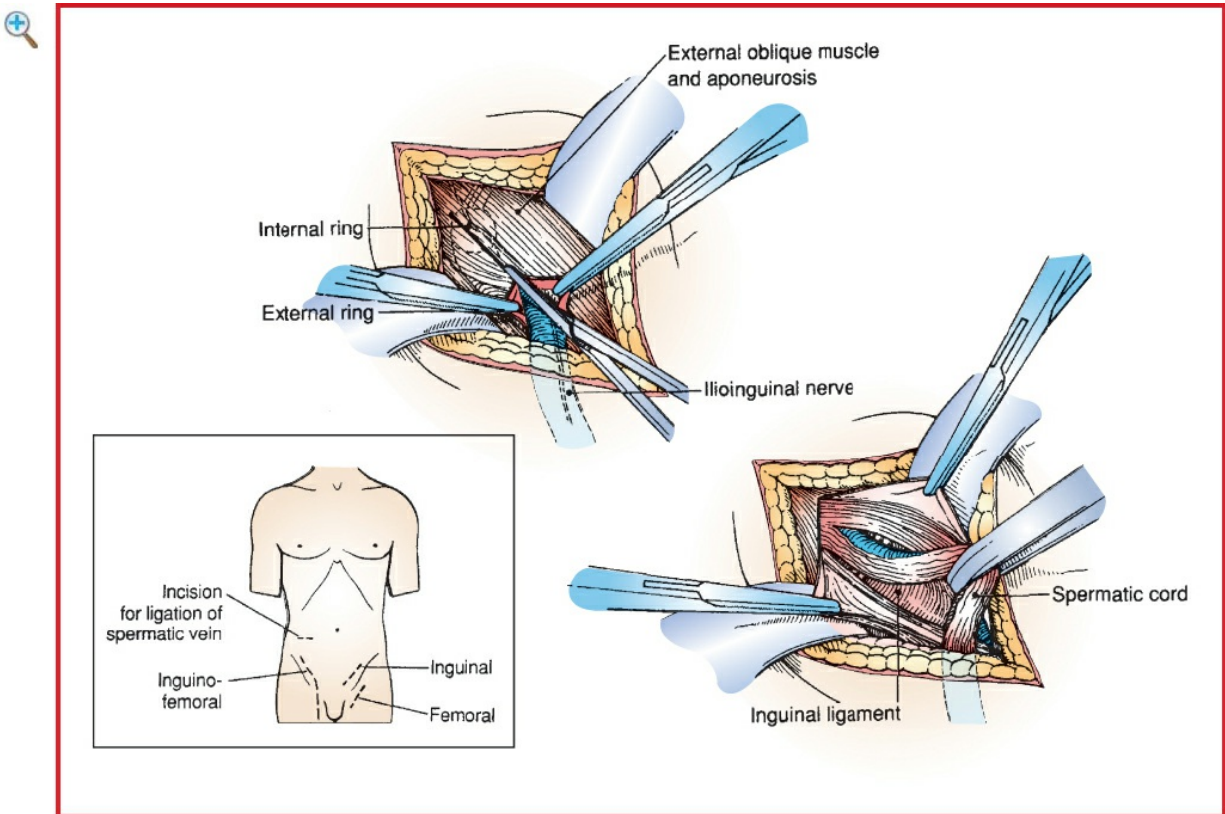


Figure 9.18 Incisions and exposure of the spermatic cord for inguinal hernia repair. (Reproduced with permission from Scott-Conner CEH, Dawson DLOperative Anatomy, 2nd edition. Lippincott Williams & Wilkins: 2003.)

Usual preop diagnosis

Varicocele causing infertility

Groin dissection, or **inguinofemoral lymphadenectomy** (lymph node dissection), is the most critical of the inguinal operations. It is performed either through an inguinal incision (Fig. 9.18, inset) curved distally over the femoral vessels or through two incisions, inguinal and upper thigh (Fig. 9.18, inset), over the femoral triangle. A complete inguinal and femoral lymphadenectomy is performed.

Usual preop diagnosis

Penile cancer

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting

NPO duration, alcohol and smoking cessation, discussion of carb loading, postop pain management options, and possibility of urinary drain postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES

	Herniorrhaphy, Orchiopexy, Orchiectomy	Ligation of Spermatic Vein	Groin Dissection
Position	Supine	⇐	⇐
Incision	Inguinal (Fig. 9.18 , inset)	Transverse groin (Fig. 9.18 , inset)	Inguinal and upper thigh (Fig. 9.18 , inset)
Antibiotics	None	⇐	⇐
Surgical time	1 h	⇐	3 h
EBL	Minimal	⇐	200 mL
Postop care	PACU → home	⇐	PACU → ward; leg elevation
Mortality	<1%	⇐	⇐
Morbidity	Wound infection: 2%	⇐	⇐
Pain score	7	5	7

■ PATIENT POPULATION CHARACTERISTICS

Age range	All ages	Young adults	Middle age
Incidence	Hernia: 5% of population Undescended testis: 0.8% of male children Testis cancer: 6/100,000	1% of young men	Extremely rare, <1% of all males
Etiology	Unknown; congenital	Varicocele	Penile cancer

(very rare)

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Typically, patients presenting for inguinal operations are healthy, with most returning home on the day of surgery. The most common inguinal operation is herniorrhaphy. In these patients, consider causes of increased intra-abdominal pressure during H&P. A hernia may strangulate → acute abdomen.

Respiratory	Chronic cough is a common precipitating factor.
Gastrointestinal	Constipation may be a precipitating factor.
Laboratory	Tests as indicated from H&P
Premedication	Midazolam 1–2 mg iv in preop area
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po. Consider aprepitant 40 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

Local anesthesia (with sedation), spinal, epidural, and GA are acceptable techniques with the choice dependent on patient age and coexisting disease, type and length of procedure, and patient preference. Local anesthesia is acceptable for simple herniorrhaphy, although discomfort may be elicited if the peritoneum is manipulated. If a spinal or epidural anesthetic is chosen, a T6 level should be sought. Most inguinal procedures are done on an outpatient basis, and the anesthetic should be planned appropriately.

Regional anesthesia

Spinal	0.75% bupivacaine 10–12 mg. For shorter procedures (<1 h), consider low-dose bupivacaine (0.75%, 7.5 mg), mepivacaine (1.5%, 45 mg), or procaine (10%, 100–150 mg). Lidocaine may be
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	used, but the incidence of transient neurologic symptoms is significant.
Epidural	1.5–2.0% lidocaine with epinephrine 5 mcg/mL, 15–25 mL; supplement with 5–10 mL as needed. Supplemental iv sedation with local or regional technique in adults, e.g., midazolam (1–2 mg iv), fentanyl (25–50 mcg iv prn anxiety or discomfort), or propofol infusion (25–50 mcg/kg/min).

General anesthesia

Induction	Standard induction. ET intubation and/or controlled ventilation may not be needed for shorter cases; consider LMA.	
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.	
Maintenance	Standard maintenance; consider propofol infusion (100–200 mcg/kg/min). Muscle relaxation is usually not required.	
Emergence	No specific considerations. PONV prophylaxis should include ondansetron.	
Blood and fluid requirements	Usually minimal blood loss IV: 18 ga × 1 BSS@ 1–2 mL/kg/h	Minimize BSS to avoid postop urinary retention after herniorrhaphy. See Zero Balance Fluid Management, Appendix K , p. K-2 and p. K-4.
Monitoring	Standard monitors	

POSTOPERATIVE

Complications	PONV Failure to void	May delay discharge from PACU → home
Multimodal pain management	Local anesthesia Ketorolac 30 mg iv in	Instillation (2 min) or infiltration of wound with

	adults ± Fentanyl 25–50 mcg iv	0.25% bupivacaine or ilioinguinal nerve block provides prolonged postop analgesia and decreases need for narcotics in outpatients. This can be used in both adult and pediatric patients.
ERAS	PONV protocol (see Appendix B, p. B-4 and B-6)	Encourage early enteral nutrition, Foley catheter removal, and mobilization.

Suggested Readings

1. Castorina S, Luca T, Privitera G, et al: An evidence-based approach for laparoscopic inguinal hernia repair: lessons learned from over 1,000 repairs. *Clin Anat* 2012; 25(6):687–96.
2. Joshi GP, Rawal N, Kehlet H, et al: Evidence-based management of postoperative pain in adults undergoing open inguinal hernia surgery. *Br J Surg* 2012; 99(2):168–85.
3. Vukovic N, Dinic L: Enhanced recovery after surgery protocols in major urologic surgery. *Front Med (Lausanne)* 2018; 5:93.

PENILE OPERATIONS

SURGICAL CONSIDERATIONS

Penectomy is the total or partial resection of the penis for squamous cell carcinoma of the penile skin. If the tumor can be resected with a safe margin of at least 2 cm, partial penectomy is usually enough. A tourniquet is placed at the base of the penis, which is amputated at least 2 cm proximal to the tumor. The corpora cavernosa are sutured, and the tourniquet is released, followed by inspection for bleeding. The edges of the urethra are sutured to the ventral skin, and the lateral and dorsal skin edges are approximated over the ends of the corpora cavernosa. Often, an **inguinal lymph node biopsy** follows the penectomy.

Usual preop diagnosis

Squamous cell carcinoma of the penile skin

Insertion of penile prosthesis is performed for impotence. The prosthesis is inserted into the corpora cavernosa ([Fig. 9.19](#)) through a penile or suprapubic incision. Penile prostheses are either malleable or inflatable. The latter have a reservoir in the retropubic space and a pump in the scrotum.

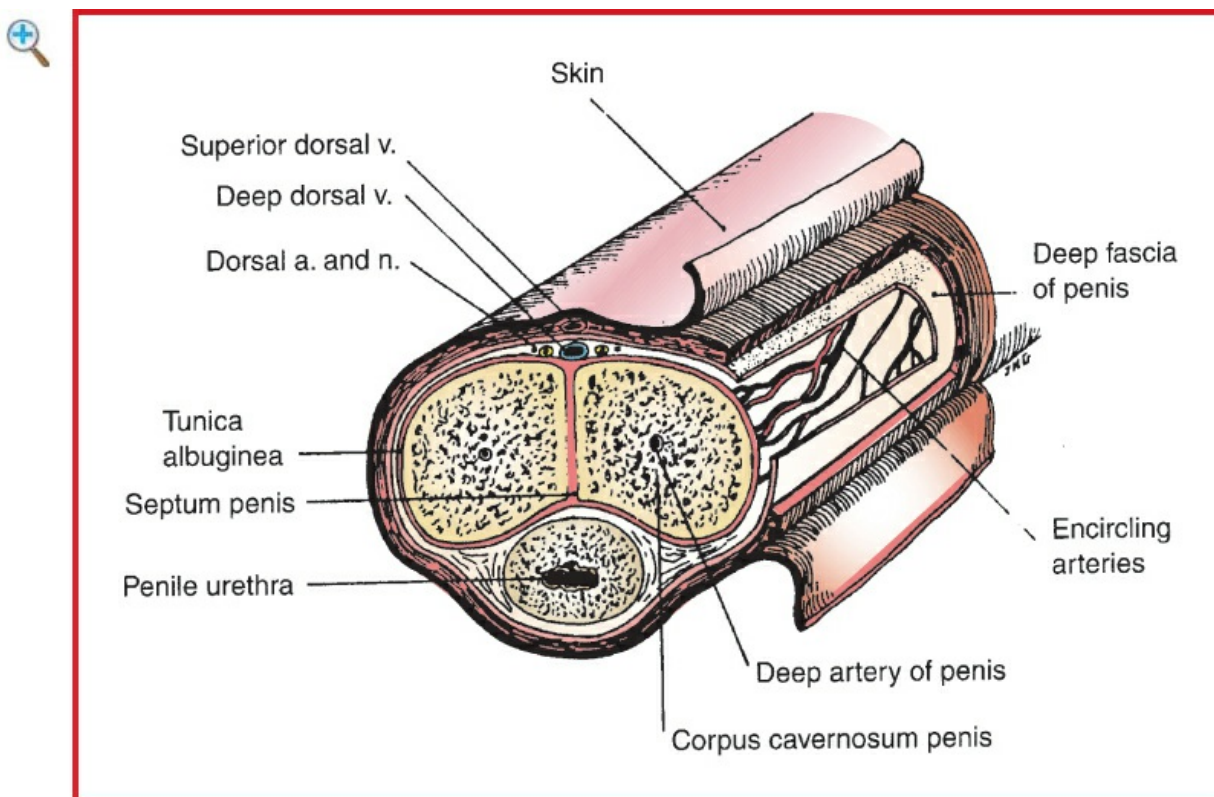


Figure 9.19 Anatomy of the penis. (Reproduced with permission from Hardy JD: Hardy's Textbook of Surgery, 2nd edition. JB Lippincott, Philadelphia: 1988.)

Usual preop diagnosis

Impotence

Hypospadias repair is performed primarily on children <5 yr.

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, postop pain management options, and possibility of urinary drain postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES

Insertion of Penile

	Penectomy	Prosthesis
Position	Supine	⇐
Incision	Circumferential penile	Bilateral incisions at the base of the penis
Special instrumentation	None	⇐
Unique considerations	None	⇐
Antibiotics	Cefazolin 2 g iv	Gentamicin 80 mg iv, slowly; ampicillin 2 g iv
Surgical time	2 h	⇐
Closing considerations	None	⇐
EBL	200 mL	⇐
Postop care	PACU → home	⇐
Mortality	<1%	⇐
Morbidity	Penile hematoma: 5%	Malfunction: 10% Edema: 5% Infection: 2% Extrusion of the prosthesis: 1%
Pain score	5	5

■ PATIENT POPULATION CHARACTERISTICS

Age range	Adults	⇐
Incidence	<1% of all males	1–2% of all males
Etiology	Poor hygiene	Organic impotence

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Patients presenting for insertion of a penile prosthesis are frequently elderly and often have preexisting medical problems, including CAD, CHF, PVD, cerebrovascular disease, COPD, and renal impairment. Preop evaluation should be directed toward the detection and treatment of these conditions prior to anesthesia.

Neurologic	Patients presenting for insertion of a penile prosthesis often have Hx of diabetes or spinal cord injury. Note the presence of neuropathy or Hx of autonomic hyperreflexia (AH) (see Anesthetic Considerations for Transurethral Procedures [except TURP]). Sx suggestive of AH include HA, flushing, nasal stuffiness, and HTN associated with voiding or noxious stimuli below the level of transection. It is important to document neurologic deficits prior to regional anesthesia.
Hematologic	Coagulation defects may be present in patients with priapism. There is a high incidence of priapism in patients with sickle cell anemia. Test: Hct, if indicated from H&P
Laboratory	Other tests as indicated from H&P
Premedication	Midazolam 1–2 mg iv in preop area
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po. Consider aprepitant 40 mg po with sip of water for GA patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

Spinal, caudal, or lumbar epidural and GA are acceptable, with choice dependent on the length of procedure, patient age, coexisting disease, and patient preference. Sacral anesthesia (saddle block) is sufficient; lumbar epidural anesthesia may be less reliable than spinal or caudal at blocking sacral fibers.

Regional anesthesia

Spinal	5% lidocaine 50 mg (controversial); 0.75% bupivacaine 10 mg in 7.5% dextrose; hyperbaric tetracaine 10 mg with epinephrine for longer procedures
Caudal	0.5% bupivacaine with epinephrine 5 mcg/mL 15–20 mL
Epidural	1.5% lidocaine with epinephrine 5 mcg/mL 15–25 mL; supplement with 5–10 mL as needed. Supplemental iv sedation

General anesthesia

Induction	Standard induction. ET intubation may not be necessary for shorter procedures; consider LMA.	
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.	
Maintenance	Standard maintenance. Deeper levels of anesthesia usually are required to obtund autonomic reflexes (e.g., HTN, laryngospasm), resulting from intense surgical stimulation that may occur during these procedures.	
Emergence	No specific considerations. Consider iv acetaminophen if po dose is given >6 h ago. PONV prophylaxis should include ondansetron.	
Blood and fluid requirements	Minimal blood loss IV: 18 ga × 1 BSS @ 1–3 mL/kg/h	See Zero Balance Fluid Management, Appendix K , p. K-2 and p. K-4.
Monitoring	Standard monitors	
Complications	AH	See Anesthetic Considerations for Transurethral Procedures.

POSTOPERATIVE

Complications	Urinary retention	
Multimodal pain management	Hydromorphone 0.2–1 mg iv or fentanyl 25–50 mcg iv prn; ketorolac 30 mg im or iv	
ERAS	PONV protocol (see Appendix B, p. B-4 and p. B-6)	Encourage early enteral nutrition, Foley catheter removal, and mobilization.

Suggested Reading

1. Vukovic N, Dinic L: Enhanced recovery after surgery protocols in major urologic surgery. *Front Med (Lausanne)* 2018; 5:93.

SCROTAL OPERATIONS

SURGICAL CONSIDERATIONS

Description

Scrotal operations are minor, common urologic procedures, performed on an outpatient basis.

Simple orchiectomy is performed as an alternative to medical castration (which uses either estrogens or LH-RH agonists on men with metastatic prostate cancer for androger ablation). Orchiectomy is always bilateral. A small scrotal incision is made, and the testis delivered. The spermatic cord is cross-clamped, transected, and suture-ligated.

Usual preop diagnosis

Metastatic prostate cancer

Vasovasostomy is the reestablishment of the continuity of the vas deferens and restoration of fertility following a previously performed vasectomy. Through a small scrotal incision, the testis and spermatic cord are delivered. The site of previous vasectomy is identified and excised and the two ends of the vas deferens are anastomosed. It is bilateral and requires the use of either the operating microscope or magnifying loupes.

Usual preop diagnosis

Infertility 2° vasectomy

Hydrocelectomy: The testis, with the surrounding hydrocele (Fig. 9.20), is delivered through a scrotal incision. The wall of the hydrocele is excised and the edges are sutured around the epididymis to prevent recurrence.

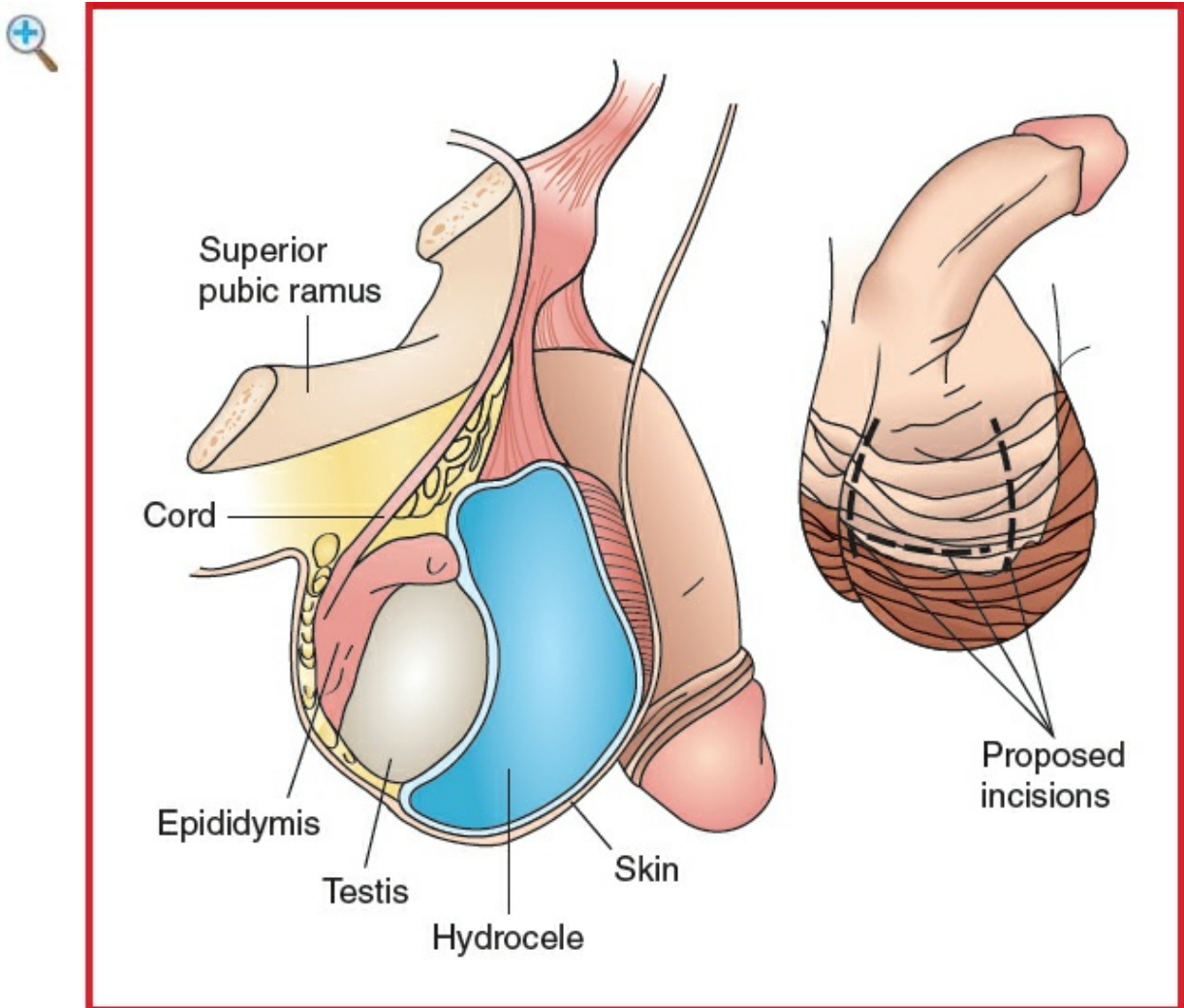


Figure 9.20 Scrotal hydrocele: scrotal incision.

Variant procedure or approach

Aspiration is used as a temporizing approach because recurrence is almost 100%.

Usual preop diagnosis

Hydrocele

Spermatocelectomy: A spermatocele is a cyst of the epididymis, usually excised with the part of the epididymis from which it arises.

Variant procedure

Aspiration as a temporizing maneuver until the operation can be performed.

Usual preop diagnosis

Spermatocele or epididymal cyst

Insertion of testicular prosthesis: A small incision is made in the scrotal skin, and a pouch is created by blunt dissection in the dartos fascia. The prosthesis is placed in the pouch and fixed to the dartos fascia to prevent prosthesis migration.

Usual preop diagnosis

Absent testis, either congenital or following orchiectomy

Reduction of testicular torsion is an emergency operation that must be performed within 6 h of occurrence to prevent irreversible ischemic damage to the testis. Through a small scrotal incision, the testis is reduced and fixed to the dartos fascia to prevent retortion.

Usual preop diagnosis

Acute testicular torsion

ERAS: For elective procedures, preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, discussion of carb loading, postop pain management options, and possibility of urinary drain postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE	
Position	Supine
Incision	Scrotal (Fig. 9.20 , inset)
Special instrumentation	Operating microscope: magnifying loupe for vasovasostomy
Antibiotics	Cefazolin 2 g iv
Surgical time	1 h
EBL	Negligible
Postop care	PACU → home
Mortality	<1%
Morbidity	Scrotal hematoma: 2% Wound infection: 2%

Pain score	4
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■ PATIENT POPULATION CHARACTERISTICS

Age range	All ages
Incidence	Common
Etiology	See preop diagnosis for each procedure above.

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Patients presenting for scrotal operations typically fall into two groups: young, otherwise healthy patients, and an older population who may present with metastatic prostate cancer accompanied by other medical conditions. The latter group is the focus of this preop evaluation.

Respiratory	Pulmonary disease may be present in elderly patients requiring orchiectomy. Tests: As indicated from H&P
Cardiovascular	Cardiac disease may be present in elderly patients requiring orchiectomy. Tests: Consider ECG; others, if indicated from H&P.
Neurologic	Document neurologic exam before regional anesthesia in patients with metastatic prostate carcinoma (spinal cord or nerve root compression may be present preop).
Musculoskeletal	✓ for the presence of spinal metastases if orchiectomy is done for palliation of prostate carcinoma. Extensive lumbar metastases may preclude the use of spinal or epidural anesthesia (relative contraindication). Tests: L-spine films if Hx is suggestive of spinal metastases.
Laboratory	Other tests as indicated from H&P
Premedication	Midazolam 1–2 mg iv in preop area

ERAS

Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po. Consider aprepitant 40 mg po with sip of water for GA patients with h/o severe PONV.

◆ INTRAOPERATIVE

Anesthetic technique

Local anesthesia (with sedation) is acceptable for simpler operations (vasectomy, orchiectomy). Procedures that are longer or more complex may require spinal, epidural, or GA. A sensory level of T10 is required to block pain 2° testicular manipulation. Many of these procedures are done on an outpatient basis, and the anesthetic should be appropriately planned to facilitate early discharge.

Regional anesthesia

Spinal	0.75% bupivacaine 10–12 mg. For shorter procedures (<1 h), consider low-dose bupivacaine (0.75%, 7.5 mg) or mepivacaine (1.5%, 45 mg). Lidocaine may be used, but the incidence of transient neurologic symptoms is significant.
Epidural	1.5–2% lidocaine with epinephrine 5 mcg/mL, 15–20 mL; supplement with 5–10 mL as needed. Supplemental iv sedation with local or regional techniques (e.g., midazolam 1–2 mg, fentanyl 25–50 mcg iv prn anxiety or discomfort).

General anesthesia

Induction	Standard induction. Consider the use of LMA.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.
Maintenance	Standard maintenance. Muscle relaxation is usually not imperative. Deeper levels of anesthesia are usually required to obtund autonomic reflexes (e.g., HTN, laryngospasm), resulting from intense surgical

	stimulation that may occur during these procedures.	
Emergence	No specific considerations. PONV prophylaxis should include ondansetron.	
Blood and fluid requirements	Minimal blood loss IV: 18 ga × 1 BSS @ 1–3 mL/kg/h	See Zero Balance Fluid Management, Appendix K , p. K-2 and p. K-4.
Monitoring	Standard monitors	
Positioning	✓ and pad pressure points ✓ eyes	NB: Peroneal nerve compression at the lateral fibular head → foot drop

POSTOPERATIVE

Complications	Peroneal nerve injury 2° lithotomy position	Peroneal nerve injury manifested by foot drop and loss of sensation on the dorsum of the foot. Seek neurology consultation.
Multimodal pain management	Acetaminophen 1 g ketorolac 30 mg im or iv in adults ± hydromorphone 0.2–1 mg iv or fentanyl 25–50 mcg iv prn	Following orchiopexy, high incidence of postop pain, which may be reduced by ilioinguinal/iliohypogastric nerve blocks
ERAS	PONV protocol (see Appendix B , p. B-4 and p. B-6)	Encourage early enteral nutrition, Foley catheter removal, and mobilization. Consider DVT prophylaxis. Chewing gum promotes return of bowel function.

Suggested Reading

1. Vukovic N, Dinic L: Enhanced recovery after surgery protocols in major urologic surgery. *Front Med (Lausanne)* 2018; 5:93.

PERINEAL OPERATIONS

SURGICAL CONSIDERATIONS

Urethroplasty: Urethral strictures that do not respond to transurethral dilation and incision are corrected with urethroplasty. A transverse or longitudinal perineal incision is made and carried down to the urethra, which is dissected free from surrounding tissues. The strictured area is excised and end-to-end anastomosis is performed over a catheter. Repair of a long urethral stricture may require placement of a patch from the scrotum, foreskin, or buccal mucosa.

Variant procedure

Transurethral incision and dilation, which is associated with a 30–50% recurrence rate.

Usual preop diagnosis

Urethral stricture, usually posttraumatic

Urethrectomy: Partial or total urethrectomy is done through a longitudinal perineal incision. The urethra is dissected free of surrounding tissues and followed proximally and distally from the membranous urethra to the external urethral meatus. In total urethrectomy, a tubularized skin graft is interposed between the membranous urethra and perineal skin.

Usual preop diagnosis

Urethral carcinoma

Insertion of artificial urinary sphincter, performed for incontinence, consists of a perineal incision, through which a cuff is inserted around the bulbar urethra. A suprapubic incision is made to place the reservoir and pump, which inflates and deflates the cuff.

Usual preop diagnosis

Urinary incontinence

Transperineal prostate seed implantation (brachytherapy)

High doses of radiation can be delivered to the prostate by implanting radioactive seeds directly into the prostate gland. Using a transrectal ultrasound probe, radioactive seeds (iodine 125 or palladium 103) are implanted into the prostate (Fig. 9.21). The patient is placed in lithotomy position, and a rectal ultrasound probe, with a perineal grid

attached, is introduced to image the prostate. Radioactive seeds are then placed transperineally using preloaded needles. Preop dosing calculations determine the number and location of the seeds. This procedure is done by a combined team of radiation oncologists and urologists. No special radiation precautions are necessary for the OR team.

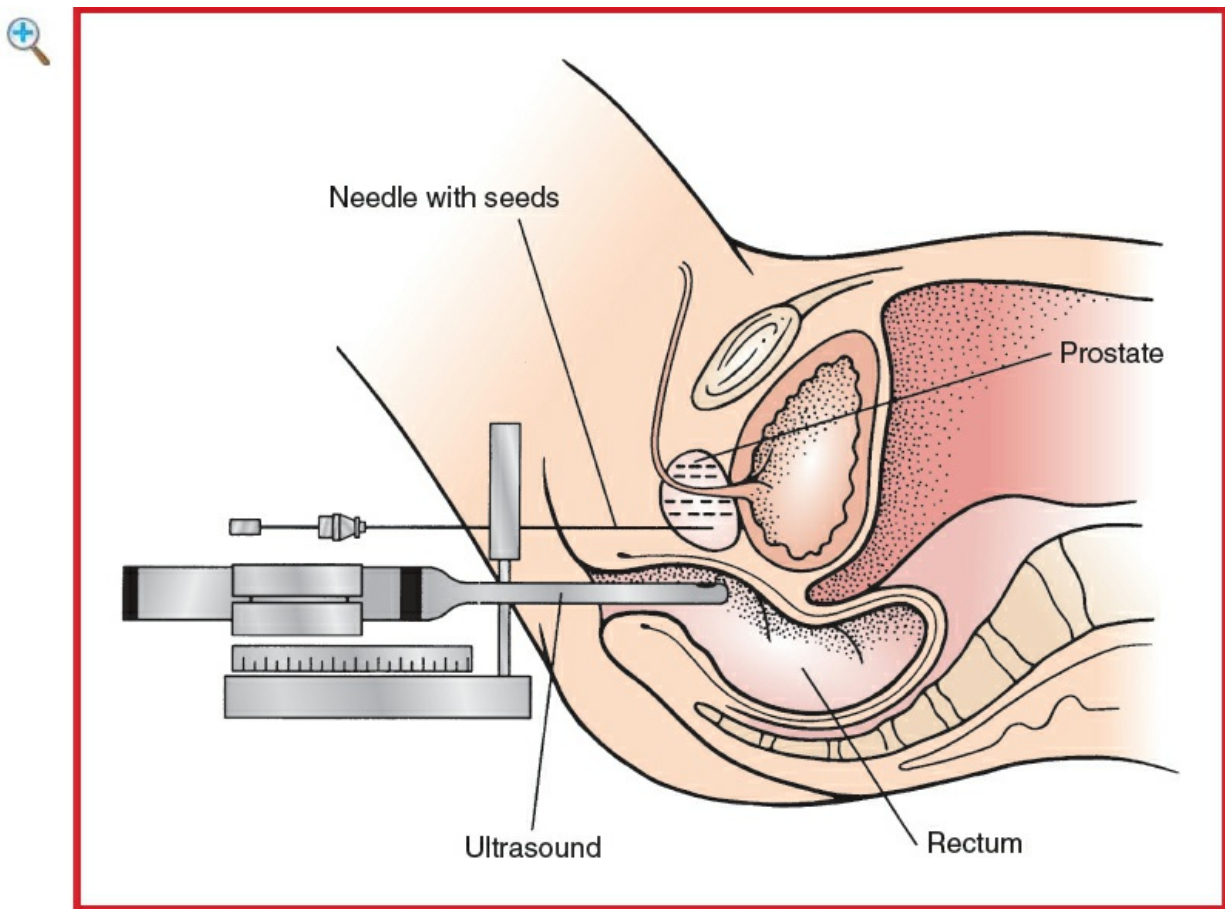


Figure 9.21 Transperineal brachytherapy of the prostate gland.

Usual preop diagnosis

Prostate cancer

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, discussion of carb loading, DVT prophylaxis, postop pain management options, and possibility of urinary drain postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES				
	Urethroplasty	Urethrectomy	Insertion of Sphincter	Brachytherapy
Position	Lithotomy	⇐	⇐	⇐
Incision	Perineal (Fig. 9.7)	⇐	⇐ and scrotal (Fig. 9.20, inset)	None
Antibiotics	Gentamicin 80 mg im/iv	⇐	⇐	⇐
Surgical time	3 h	2 h	3 h	2 h
EBL	100 mL	300 mL	Minimal	⇐
Postop care	PACU → home	⇐	⇐	PACU, outpatient
Mortality	<1%	⇐	⇐	⇐
Morbidity	Wound infection: 2%	⇐	⇐ Erosion of the urethra: 10% Extrusion of sphincter: 2%	Urinary retention
Pain score	3	3	4	3

■ PATIENT POPULATION CHARACTERISTICS				
Age range	All ages	Adults	Older adults	50–80 yr
Incidence	<1% of urologic procedures	⇐	2% of radical prostatectomy	10% of prostate cancer
Etiology	Traumatic strictures	Unknown	Radical prostatectomy (2%); incontinence	Aging

■ ANESTHETIC CONSIDERATIONS

■ PREOPERATIVE

This is a generally healthy patient population; preop considerations should be based on H&P.

Laboratory	Tests as indicated from H&P
Premedication	Sedation prn anxiety in adults (e.g., lorazepam 1–2 mg po 1–2 h prior to surgery; midazolam 1–2 mg iv in preop area)
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg. Consider aprepitant 40 mg po with sip of water for GA patients

with h/o severe PONV.

◆ INTRAOPERATIVE

Anesthetic technique

Spinal or GA is acceptable, with choice dependent on the length of procedure, position, patient age, coexisting disease, and patient preference. A sacral sensory level (saddle block) is usually sufficient. Lumbar epidural anesthesia may be less reliable at providing sacral anesthesia and offers no advantages over the above techniques for shorter procedures, although caudal anesthesia may be an acceptable alternative.

Regional anesthesia

Spinal	0.75% bupivacaine 10 mg in 7.5% dextrose: hyperbaric tetracaine 10 mg (with epinephrine [200 mcg] for longer procedures)
Caudal	0.5% bupivacaine with epinephrine 5 mcg/mL 15–20 mL. Supplemental iv sedation

General anesthesia

Induction	Standard induction. Consider the use of LMA.	
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.	
Maintenance	Standard maintenance; muscle relaxation is usually not required. Deeper levels of anesthesia are usually required to obtund autonomic reflexes (e.g., HTN, laryngospasm), resulting from intense surgical stimulation that may occur during these procedures.	
Emergence	No specific considerations. PONV prophylaxis should include ondansetron.	
Blood and fluid requirements	Minimal blood loss IV: 20–18 ga × 1 BSS @ 1–3 mL/kg/h	See Zero Balance Fluid Management, Appendix K , p. K-2 and p. K-4.
Monitoring	Standard monitors	

Positioning	✓ and pad pressure points ✓ eyes	Patients with arthritis or other musculoskeletal disorders may not tolerate the exaggerated lithotomy position, thus precluding the use of a regional technique. NB: In lithotomy position, peroneal nerve compression at the lateral fibular head → foot drop.
Complications	Anticipate ↓ BP on return from lithotomy position.	Rx: volume (200–500 mL NS/LR) or ephedrine (5 mg iv) may be necessary.

▼ POSTOPERATIVE

Complications	Peroneal nerve injury 2° lithotomy position	Peroneal nerve injury manifested as foot drop with loss of sensation over the dorsum of the foot. Seek neurology consultation.
Multimodal pain management	Emphasize multimodal approach combining nonopioid and bolus opioid analgesics (see Appendix K, p. K-6).	Acetaminophen 1 g po or iv ± Hydromorphone 0.2–1 mg iv prn ± Ketorolac or celecoxib
ERAS	PONV protocol (see Appendix B, p. B-4 and p. B-6)	Encourage early enteral nutrition, Foley catheter removal, and mobilization.

Suggested Reading

1. Vukovic N, Dinic L: Enhanced recovery after surgery protocols in major urologic surgery. *Front Med (Lausanne)* 2018; 5:93.

VAGINAL OPERATIONS

SURGICAL CONSIDERATIONS

Description

Vaginal operations are performed by both urologists and gynecologists. They include the following:

Repair of vesicovaginal fistulas: The vaginal approach is usually recommended for small and distally located vesicovaginal fistulas; otherwise, a transabdominal repair is performed (see Open Bladder Operations, [p. 1002](#)). An incision is made in the anterior vaginal wall around the fistula, which is excised. Bladder and vaginal walls are separated and closed with interposition of tissues or flaps to separate the incisions and prevent recurrence. A Foley catheter is left indwelling.

Variant approach

Transabdominal repair of vesicovaginal fistula (see Open Bladder Operations, [p. 1002](#)).

Usual preop diagnosis

Vesicovaginal fistula

Operations to correct stress urinary incontinence: Many procedures have been designed to correct female urinary incontinence. They fall into two basic groups: (a) operations to correct hypermobility of the urethra and (b) operations to correct the nonfunctioning urethra. The operation most commonly used by urologists to correct hypermobility is the **Stamey procedure** ([Fig. 9.22](#)) or **vesical neck suspension**. The operation is performed through two small suprapubic incisions, one on each side of the midline, and an anterior vaginal incision. A nylon suture is placed in a loop from either side of the bladder neck and not around it. Cystoscopy is used to ensure proper placement and to prevent the suture from transversing the bladder. When the sutures are pulled up and tied over the anterior rectus sheath, they pull the bladder neck up to its original position behind the symphysis pubis and restore the acute posterior ureterovesical angle. A variant of this procedure is the **Raz bladder neck suspension**, where bolsters are not used.

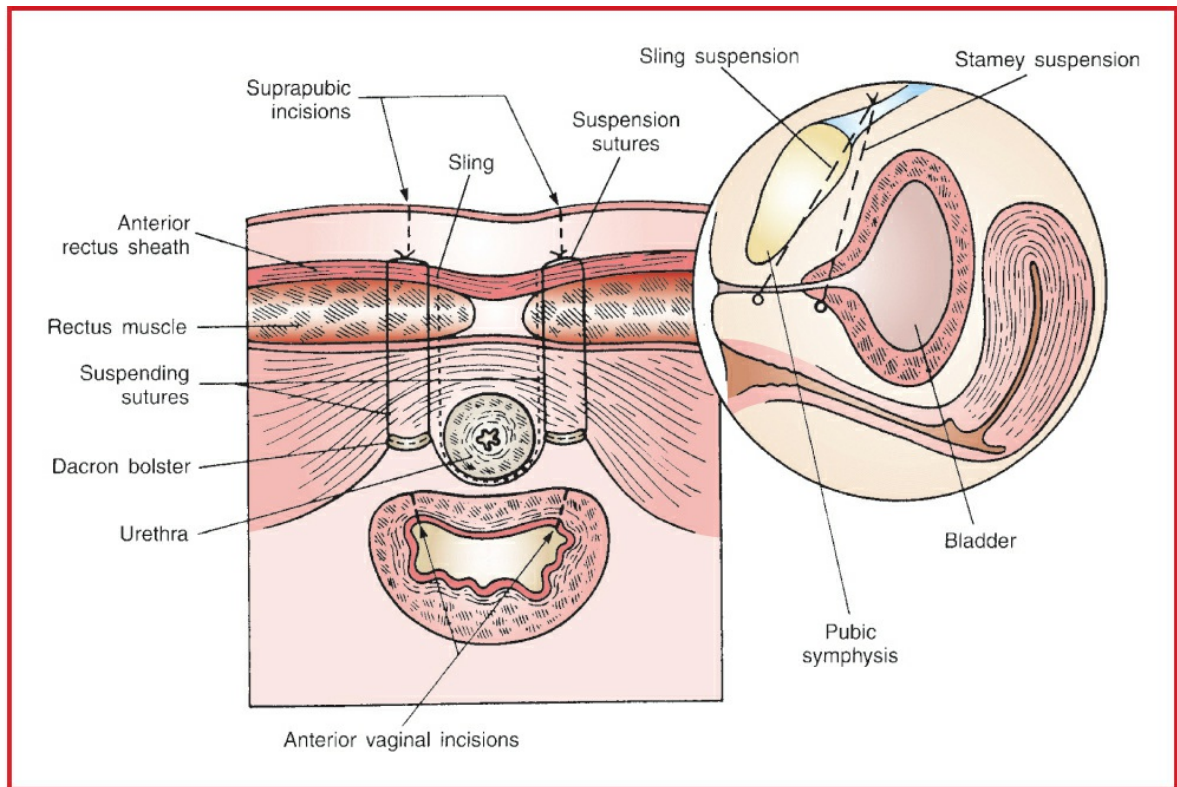


Figure 9.22 Stamey and sling procedures, sectional view; **inset** shows lateral view.

Operations to correct a nonfunctioning urethra include **submucosal collagen injection** at the bladder neck or construction of a **sling**. The rectus fascia, the fascia lata of the thigh, or the vaginal wall can be used to construct a sling around the urethra. All these techniques involve a combined suprapubic and vaginal approach.

Sling operations are the most common procedures done for correction of stress incontinence. These procedures can be done vaginally or open via a suprapubic incision. A variety of materials, both natural and synthetic, have been used. Many modifications have been made with reference to the placement of anchorage of the sling; therefore, there are a large number of procedures with different names utilizing the same principle. Most slings are placed midurethra.

Variant approach

The **Marshall-Marchetti-Krantz operation**, which is performed retropubically, sutures the anterior portion of the urethra, bladder neck, and bladder to the pubic bone. Other modifications include laparoscopic suspensions and ligament surgery. Some patients have cystoceles and rectoceles, which can be repaired at the same time as the vaginal sling surgery.

Usual preop diagnosis

Stress urinary incontinence

Excision of urethral diverticulum: Urethral diverticula are extremely rare and need excision only if they are the cause of recurrent UTIs. An incision is made in the anterior vaginal wall over the urethral diverticulum, which is dissected all around until it is attached only by its neck. It is excised and the neck closed. A Foley catheter is left indwelling, and the vaginal incision is closed.

Usual preop diagnosis

Recurrent UTI 2° infected urethral diverticulum

Repair of cystocele and rectocele: Some patients with urinary incontinence also present with prolapse of the bladder or rectum into the vagina. These can be repaired at the same time as incontinence surgery. A vaginal incision (anterior for cystocele, posterior for rectocele) is made and dissected laterally to free the bladder or rectum from the vagina. The defect is repaired and the redundant vaginal wall excised.

Usual preop diagnosis

Incontinence with pelvic prolapse, cystocele, or rectocele

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, discussion of carb loading, DV1 prophylaxis, postop pain management options, and possibility of urinary drain postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES			
	Repair of Vesicovaginal Fistula	Correction of Stress Incontinence	Excision of Urethral Diverticulum
Position	Lithotomy	⇐	⇐
Incision	Anterior vaginal	Anterior vaginal; suprapubic	Anterior vaginal
Special instrumentation	None	Cystoscope	None
Antibiotics	Gentamicin 80 mg iv, slowly	⇐	⇐
Surgical time	2 h	1 h	⇐

EBL	200 mL	500 mL	200 mL
Postop care	PACU → room	⇐	⇐
Mortality	<1%	⇐	⇐
Morbidity	Infection: 2% Recurrence: 2%	⇐ 10% Urinary retention: 20%	⇐ ⇐
Pain score	3	5	3

■ PATIENT POPULATION CHARACTERISTICS

Age range	20–80 yr	⇐	⇐
Incidence	<1% of urologic procedures	5% of urologic procedures	<1% of urologic procedures
Etiology	Traumatic delivery; iatrogenic following hysterectomy (<1%)	Childbirth	Congenital (extremely rare)

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

This is a generally healthy patient population. Preop considerations should be based on H&P.

Laboratory	Tests as indicated from H&P
Premedication	Standard premedication
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po. Consider aprepitant 40 mg po with sip of water for GA patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

Spinal, continuous lumbar epidural, or GA are acceptable, with choice dependent on age, coexisting disease, and patient preference. A block level of T9-T10 is recommended for operations involving the bladder, whereas somewhat higher levels of anesthesia may be necessary if a suprapubic incision is made. Epidural anesthesia may be less reliable than spinal in providing sacral anesthesia.

Regional anesthesia

Spinal	0.75% bupivacaine 10–12 mg (1.6 mL)
Epidural	2% lidocaine with epinephrine 5 mcg/mL, 15–20 mL. Supplemental iv sedation

General anesthesia

Induction	Standard induction	
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.	
Maintenance	Standard maintenance. Muscle relaxation is not imperative.	
Emergence	No specific considerations. PONV prophylaxis should include ondansetron.	
Blood and fluid requirements	Minimal blood loss IV: 18 ga × 1 BSS @ 1–3 mL/kg/h	See Zero Balance Fluid Management, Appendix K , p. K-2 and p. K-4.
Monitoring	Standard monitors	
Positioning	✓ and pad pressure points ✓ eyes	NB: In lithotomy position, peroneal nerve compression at the lateral fibular head → foot drop
Complications	Anticipate ↓ BP when returning from lithotomy position. Bladder perforation	Rx: volume (200–500 mL NS/LR) or ephedrine (5 mg iv) may be necessary. Bladder perforation may present as shoulder pain

in the awake patient but may go unnoticed in the patient under GA. Sx include unexplained HTN, tachycardia, and ↓ BP (rare).

▼ POSTOPERATIVE

Complications	Peroneal nerve injury 2° lithotomy position Bladder perforation (see above)	Peroneal nerve injury is manifested as foot drop with loss of sensation on the dorsum of the foot. Seek neurology consultation.
Multimodal pain management	Regional analgesia when possible Acetaminophen 1 g po or iv	Consider ketorolac 30 mg iv in adults; supplement with hydromorphone 0.2–1 mg iv prn.
ERAS	PONV protocol (see Appendix B, p. B-4 and p. B-6)	Encourage early enteral nutrition, Foley catheter removal, and mobilization. Consider DVT prophylaxis.

Suggested Readings

1. Ashok K, Petri E: Failures and complications in pelvic floor surgery. *World J Urol* 2012; 30(4):487–94.
2. Vukovic N, Dinic L: Enhanced recovery after surgery protocols in major urologic surgery. *Front Med (Lausanne)* 2018; 5:93.

SPECIAL CONSIDERATIONS FOR ROBOTIC-ASSISTED LAPAROSCOPIC PROCEDURES

■ ANESTHETIC CONSIDERATIONS

The anesthetic considerations for the majority of laparoscopic and robotic procedures on the urologic system are similar and are thus summarized in this section rather than repeated for each individual procedure.

INTRAOPERATIVE

Anesthetic technique

In general, because laparoscopic surgery involves insufflation of the abdomen with CO₂ and may also require steep Trendelenburg position, GA with tracheal intubation and controlled ventilation is the technique of choice.

General anesthesia

Induction	Standard induction	
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.	
Maintenance	Standard maintenance. Muscle relaxation is advised.	
Emergence	Ensure airway patency (subcutaneous emphysema → airway compromise). Consider iv acetaminophen if po dose given >6 h ago. PONV prophylaxis should include ondansetron.	
Blood and fluid requirements	IV: 16–18 ga × 1 BSS @ 1–3 mL/kg/h Warm fluid and humidify gases	Blood loss may be difficult to estimate. In general, less than for open procedures. See Zero Balance Fluid Management, Appendix K , p. K-2 and p. K-4.
Monitoring	Standard monitors ± Arterial line ± CVP	Consider invasive monitors for nephrectomy and prostatectomy.
Positioning	✓ and pad pressure points ✓ eyes	Prolonged duration of robotic procedures in lithotomy position of special concern (peroneal nerve injury)

Complications	CO ₂ insufflation	→ ↑ intra-abdominal pressure → ↓ venous return, ↓ BP; also ↑ airway pressure, ↑ PCO₂/ETCO₂ NB: Right mainstem intubation can occur because of shift of diaphragm cephalad.
	Retroperitoneal gas insufflation VAE	→ subcutaneous emphysema (can track to the head and neck → airway compromise), pneumothorax Gas embolism via prostatic venous sinuses reported

▼ POSTOPERATIVE

Complications	Peroneal nerve injury (foot drop) Upper airway obstruction	Other pressure point injuries may occur 2° subcutaneous emphysema tracking up to the head and neck.
Multimodal pain management	Consider acetaminophen ± Ketorolac 30 mg iv/im in adults	Pain less than for open procedures. Supplement with hydromorphone 0.2–1 mg iv prn followed by PCA (see Appendix K, p. K-6).
ERAS		Encourage early enteral nutrition, Foley catheter removal, and mobilization. Consider DVT prophylaxis. Chewing

gum promotes return of
bowel function.

Test

Hct

If excessive blood loss

Suggested Readings

1. Conacher ID, Soomro NA, Rix D: Anaesthesia for laparoscopic urological surgery Br J Anaesth 2004; 93(6):859–64.
2. Nelson JB: Debate: open radical prostatectomy vs. laparoscopic vs. robotic. Urol Oncol 2007; 25(6):490–3.
3. Zorn KC, Gofrit ON, Orvieto MA, et al: Da Vinci robot error and failure rates: single institution experience on a single three-arm robot unit of more than 700 consecutive robot-assisted laparoscopic radical prostatectomies. J Endourol 2007; 21(11):1341–4.

Suggested Viewing

Links are available online to the following videos:

TURP: <https://www.youtube.com/watch?v=WNCsunVC4Rs>

KTP Laser TURP (green light laser prostatectomy): <https://www.youtube.com/watch?v=TKtMMao5vpY>

Weill Cornell Robotic Prostatectomy: 3-D Animation (2/6): <https://www.youtube.com/watch?v=5Exd86dbT68>

Nephrectomy (laparoscopic, hand-assisted): https://www.youtube.com/watch?v=JJK5n6RGC_s

Robotic / DaVinci™ Partial Nephrectomy; Kurt Strom MD, Joshua Holyoak MI
<https://www.youtube.com/watch?v=w5zfOsyzlWQ>

Minimally Invasive, Open Robotic Cystectomy for Bladder Cancer Removal
<https://www.youtube.com/watch?v=fJ7JjDqrr-g>